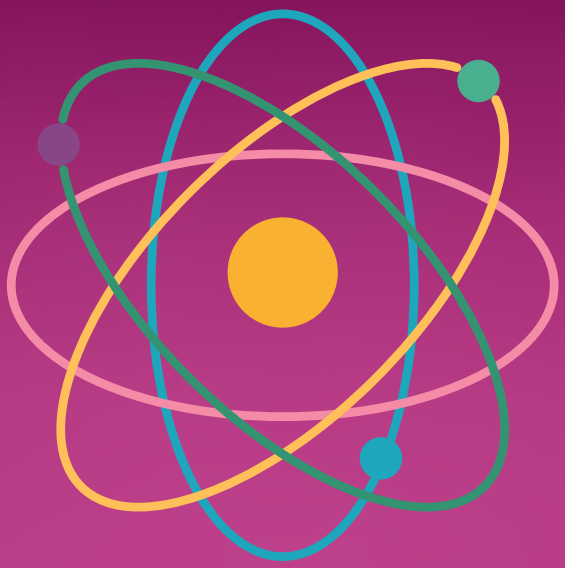
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# PHYSICAL CHEMISTRY

## JEE (MAIN+ADVANCED)

ENTHUSIAST COURSE

**EXERCISE**

## Electrochemistry

English Medium

## EXERCISE # S-I

***Note : If not given in question Use :*** $\frac{2.303RT}{F} = 0.059$

***Electrode potential cell emf.***

**1.** For the cell reaction $2Ce^{4+} + Co \longrightarrow 2Ce^{3+} + Co^{2+}$

$E^o_{cell}$ is 1.89 V. If $E^o_{Co^{2+}|Co}$ is – 0.28 V, what is the value of $E^o_{Ce^{4+}|Ce^{3+}}$ in volts ?

**EC0239**

**2.** Determine the standard reduction potential in volts for the half reaction :

$Cl_2 + 2e^- \longrightarrow 2Cl^-$

Given $Pt^{2+} + 2Cl^- \longrightarrow Pt + Cl_2$, $E^o_{Cell} = -0.15$ V

$Pt^{2+} + 2e^- \longrightarrow Pt$ $E^\circ = 1.20$ V

**EC0193**

**3.** If $E^o_{Fe^{2+}|Fe} = -0.44$ V, $E^o_{Fe^{3+}|Fe^{2+}} = 0.77$ V. Calculate $E^o_{Fe^{3+}|Fe}$ in volts.

**EC0005**

**4.** If for the half cell reactions

$Cu^{2+} + e^- \longrightarrow Cu^+$ $E^\circ = 0.15$ V

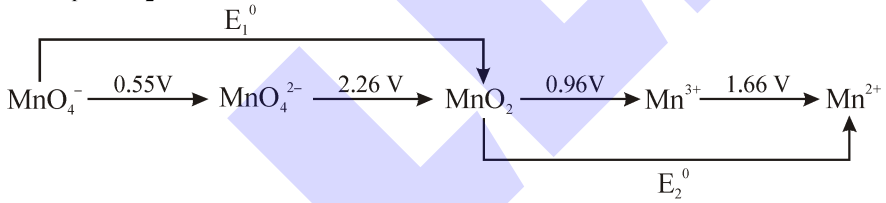
$Cu^{2+} + 2e^- \longrightarrow Cu$ $E^\circ = 0.34$ V

Calculate $E^\circ$ in volts of the half cell reaction

$Cu^+ + e^- \longrightarrow Cu$

and calculate $E^0$ in volts for disproportion of $Cu^+$ **EC0239**

**5.** From the standard potential in acidic medium as shown in the following latimer diagram , the value of $(E_1^o + E_2^o)$, in volts, is - **EC0013**




**6.** Find $E^0$ at 298 K for the reaction in volts, **EC0240**

$Cu^{2+}(aq) + In^{2+}(aq) \rightleftharpoons Cu^+(aq) + In^{3+}(aq)$

Given that $E^\circ_{Cu^{2+}|Cu^+} = 0.15V$ , $E^\circ_{In^{3+}|In^+} = -0.42V$ , $E^\circ_{In^{2+}|In^+} = -0.40V$

**7.** The half cell reactions for rusting of iron are: **EC0241**

$2H^+ + \frac{1}{2}O_2 + 2e^- \longrightarrow H_2O$; $E^0 = +1.23$ V, $Fe^{2+} + 2e^- \longrightarrow Fe$; $E^0 = -0.44$ V

Calculate $\Delta G^0$ (in kJ) for rusting of 1 mole of iron (Give your answer as nearest integer).

**8.** For the reaction, $4Al(s) + 3O_2(g) + 6H_2O + 4\ OH^- \rightleftharpoons 4\ [Al(OH)_4^-]$ ; $E^\circ_{cell} = 2.73$ V.

If $\Delta G^\circ_f(OH^-) = -157$ kJ mol$^{-1}$ and $\Delta G^\circ_f(H_2O) = -237.2$ kJ mol$^{-1}$, determine $\Delta G^\circ_f[Al(OH)_4^-]$ (in kJ/mole) [Given : $96.5 \times 2.73 = 263.445$]

(Give your answer as nearest integer).

**EC0199**

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**9.** For a cell,
$4B(s) + 3O_2(g) \rightarrow 2B_2O_3(g)$ ; $E^o_{cell} = 1.50$ volts
What is molar entropy (in $JK^{-1}mol^{-1}$) of oxygen gas at 300K ? **EC0242**

Given : $\Delta H^0_f [B_2O_3, g] = -840\ kJ\,mol^{-1}$

$S^0_m [B_2O_3, g] = 280\ JK^{-1}\,mol^{-1}$

$S^0_m [B, s] = 10\ JK^{-1}\,mol^{-1}$

**10.** A zinc electrode is placed in a 0.1M ZnX solution at 25°C. Assuming that the salt (ZnX) is 20% dissociated at this concentration calculate the electrode reduction potential in volts .
$E^0 (Zn^{2+}| Zn) = -0.76V$.
**EC0010**

**11.** The reduction potential of hydrogen electrode when placed in a buffer solution is found to be – 0.413V. The pH of the buffer solution is -
**EC0014**

**12.** At $2.97 \times 10^{-x}$ M concentration of $Cu^{2+}$ in a solution of $CuSO_4$ , electrode potential of copper is zero at 25°C then x will be - Given : $E^0 (Cu \mid Cu^{2+}) = -0.34$ V. $[10^{-11.525} = 2.97 \times 10^{-12}]$
**EC0243**

**13.** Calculate the EMF of a Daniel cell (in volts) when the concentration of $ZnSO_4$ and $CuSO_4$ are 0.001 M and 0.1M respectively. The standard potential of the cell is 1.1V.
**EC0006**

**14.** Calculate $E^0$ and E (in volts) for the cell $Sn \mid Sn^{2+}(1M) \parallel Pb^{2+}(10^{-3}M) \mid Pb$, $E^0 (Sn^{2+}| Sn) = -0.14V$, $E^0 (Pb^{2+}| Pb) = -0.13V$ ? **EC0007**

**15.** Consider a Galvanic cell, **EC0244**

$$Zn(s) \mid Zn^{2+}(0.1\ M) \parallel Cu^{2+}(0.1\ M) \mid Cu(s)$$

by what factor, the electrolyte solution in anodic half cell should be diluted to increase the emf by 9 milli volts at 298 K. $\left( \text{Given} : \frac{2.303RT}{F} = 0.06 \text{ and } \log 2 = 0.3 \right)$

**16.** Calculate potential in volts of reaction given below : **EC0245**
$2Ag^+(1M) + H_2(1bar) \rightarrow 2Ag(s) + 2H^+(1M)$

$E^o_{Ag^+|Ag} = 0.80$ V, $E^o_{H^+|H_2(Pt)} = 0.0$ V

**17.** The $pK_{sp}$ of AgI is 16.07 . If the $E^o$ value for $Ag^+| Ag$ is 0.7991 V . Find the $E^o$ in volts for the half cell reaction $AgI(s) + e^- \longrightarrow Ag + I^-$. $[16.07 \times 0.059 = 0.94813]$
**EC0009**

**18.** Calculate standard electrode reduction potential in volts when silver wire coated with a layer of silver iodide is placed in solution of KI at 25ºC.

Given : $E^o_{Ag^+ / Ag} = 0.8$ V ; $K_{sp}(AgI) = 10^{-16}$ ; $\frac{2.303\ RT}{F} = 0.06$ **EC0246**

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## EQUILIBRIUM CONSTANT

**19.** If value of equilibrium constant for given reaction is $A \times 10^{12}$ then value of A will be : **EC0247**

$Fe^{2+} + Ce^{4+} \rightleftharpoons Fe^{3+} + Ce^{3+}$, [Given : $E^0_{Ce^{4+}|Ce^{3+}} = 1.44$ V; $E^0_{Fe^{3+}|Fe^{2+}} = 0.68$ V ] [$10^{12.88} = 7.6 \times 10^{12}$]

**20.** If value of equilibrium constant is $A \times 10^{25}$ then value of A will be for the reaction : **EC0248**

$Fe + CuSO_4 \rightleftharpoons FeSO_4 + Cu$ at 25°C.

Given $E^0$ $(Fe|Fe^{2+}) = 0.44V$, $E^0$ $(Cu|Cu^{2+}) = -0.337V$. [$10^{26.3389} = 2.18 \times 10^{26}$]

**21.** At $25^0$C the value of equilibrium constant for reaction $Fe^{3+} + Ag \rightleftharpoons Fe^{2+} + Ag^+$ is 0.531M. The standard electrode potential for $Ag^+ + e^- \rightleftharpoons Ag$ is 0.799V. What is the standard potential in volts for $Fe^{3+} + e^- \rightleftharpoons Fe^{2+}$ ? [log (0.531 = – 0.27)]

**EC0197**

**22.** The standard reduction potential at 25°C for the reduction of water **EC0249**

$2H_2O + 2e^- \rightleftharpoons H_2 + 2OH^-$ is –0.8277 volt. Calculate nearest integer value of $pK_W$ for reaction.

$2H_2O \rightleftharpoons H_3O^+ + OH^-$ at 25°C. [193 × 0.8277 = 159.75] [2.303 RT = 5705.35 J/mol]

**23.** For a cell Mg(s) | $Mg^{2+}$(aq) || $Ag^+$ (aq) | Ag, **EC0250**

(i) If equilibrium constant at 25°C is $10^x$ then nearest integer value of x is.

(ii) Also find the maximum work (in kJ) per mole of Mg that can be obtained by operating the cell in standard conditions.

$E^0_{(Mg^{2+}|Mg)} = -2.3565V$, $E^0_{(Ag+|Ag)} = 0.8$ V.

**24.** For the cell reaction : $Hg_2Cl_2(s) + 2Ag(s) \rightarrow 2Hg(l) + 2AgCl(s)$

temperature coefficient of cell emf is found to be 0.02 $VK^{-1}$. Find $\Delta_r S^o$ for cell reaction in kJ $mole^{-1}$

**EC0012**

## CONCENTRATION CELLS :

**25.** Calculate the EMF of the following cell in volts at 298K.

$Zn | Zn^{2+}$ (0.01M) || $Zn^{2+}$ (0.1 M) | Zn

**EC0015**

**26.** Calculate pH using the following cell :

Pt ($H_2$) | $H^+$ (x M) | | $H^+$ (1 M) | Pt ($H_2$) if $E_{cell} = 0.2364$ V.

1 atm 1 atm

**EC0016**

**27.** Equinormal Solutions of two weak acids, HA ($pK_a = 3$) and HB ($pK_a = 5$) are each placed in contact with equal pressure of hydrogen electrode at 25°C. When a cell is constructed by interconnecting them through a salt bridge, find the emf of the cell.(in volts)

**EC0200**

**28.** In two vessels each containing 500ml water, 0.5m mol of aniline ($K_b = 10^{-9}$) and 50mmol of HCl are added separately. Two hydrogen electrodes are constructed using these solutions. Calculate the emf in volts of cell made by connecting them appropriately.

**EC0201**

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## ELECTROLYTIC CELL :

**29.** Calculate mole of electrons lost or gained during electrolysis of **EC0251**
(a) 3.55 gm of $Cl^-$ ions (b) 1 gm $Cu^{2+}$ ions (c) 2.7 gm of $Al^{3+}$ ions
[Atomic mass : Cu = 63.5 , Al = 27, Cl = 35.5]

**30.** How many faradays of electricity are involved in each of the case

(a) 0.25 mole $Al^{3+}$ is converted to Al.

(b) 27.6 gm of $SO_3$ is convered to $SO_3^{2-}$

(c) The $Cu^{2+}$ in 1100 ml of 0.5 M $Cu^{2+}$ is converted to Cu.

**EC0018**

**31.** Chromium metal can be plated out from an acidic solution containing $CrO_3$ according to following equation :

$CrO_3(aq) + 6H^+(aq) + 6e^- \rightarrow Cr(s) + 3H_2O$

Calculate :

(i) How many grams of chromium will be plated out by 24125 coulombs and

(ii) How much time in seconds will it take to plate out 1.5 gm of chromium by using 12.5 ampere current ? [Atomic mass : Cr = 52]

**EC0202**

**32.** During electrolysis of NaCl, if 3 moles of NaCl are electrolysed then how much charge in faradays is required if current efficiency is 75% for the process? **EC0252**

**33.** A constant current of 30 A is passed through an aqueous solution of NaCl for a time of 1.0 hr. How many grams of NaOH are produced? What is volume of $Cl_2$ gas at 1atm,273K produced (in litre) ?

**EC0021**

**34.** A current of 0.01 amperes is passed through a solution of a metal bromide ($MBr_x$). The only reaction at the cathode is the deposition of metal (M). After 3 hours, 0.072 gm of metal (M) is deposited. The value of 'x' is

(Given : Atomic wt. of M = 192) **EC0253**

**35.** If during electrolysis of a water sample 0.224 litre of $H_2$ gas is formed at the cathode, how much volume of $O_2$ gas (in litres) is formed at the anode under identical conditions ? **EC0254**

**36.** 0.5 mole of electron is passed through two electrolytic cells in series. One contains silver ions, and the other zinc ions. Assume that only cathode reaction in each cell is the reduction of the ion to the metal. How many grams of each metals will be deposited ? **[At weight Zn = 65, Ag = 108]**

**EC0019**

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**37.** The electrosynthesis of $MnO_2$ is carried out from a solution of $MnSO_4$ in $H_2SO_4$ (aq). If a current of 25.5 ampere is used with a current efficiency of 85%, to produce 1 kg of $MnO_2$ , B × $10^5$seconds time is required then value of B will be ? [$\frac{965}{87\times25.5\times85}=5.12\times10^{-3}$] [Atomic mass : Mn = 55]. **EC0255**

**38.** Calculate the quantity of electricity (in coulomb) that would be required to reduce 12.3 g of nitrobenzene to aniline, if the current efficiency for the process is 50 percent. If the potential drop across the cell is 3.0 volts, how much energy (in kJ) will be consumed? **EC0256**

**39.** Assume 96500 C as one unit of electricity. If cost of electricity of producing 10 gm Al is Rs 10, what is the cost of electricity of producing 10 gm Mg in Rupees ?

**EC0023**

**40.** A metal (M) is known to form fluoride $MF_2$. When 10A of electricity is passed through a molten salt for 330 sec., 1.95g of metal is deposited. Find the atomic weight of M. What will be the quantity of electricity (in coulomb) required to deposit the same mass of Cu from $CuSO_4$?

$\left[\frac{1.95\times965}{63.5}=29.63,\frac{1.95\times965}{33}=57\right]$ **EC0257**

**41.** After electrolysis of NaCl solution with inert electrodes for a certain period of time. 600 mL of the solution was left. Which was found to be 1N in NaOH. During the same time, 31.75 g of Cu was deposited in the copper voltameter in series with the NaCl electrolytic cell. Calculate the percentage yield of NaOH obtained ?

**EC0204**

**42.** A solution of $Ni(NO_3)_2$ is electrolysed between platinum electrodes using a current of 5 ampere for 20 mintue. What mass of Ni (in grams) is deposited at the cathode?

**EC0025**

**43.** A current of 3.7A is passed for 6hrs. between Ni electrodes in 0.5L of 2M solution of $Ni(NO_3)_2$. What will be the molarity of $Ni(NO_3)_2$ solution at the end of electrolysis?

**EC0026**

## CONDUCTANCE

**44.** A solution containing 2.08 g of anhydrous barium chloride in 500 $cm^3$ of water has a specific conductivity 0.005 $ohm^{-1}cm^{-1}$. What are molar and equivalent conductivities of this solution in mho $cm^2\,mol^{-1}$ and mho $cm^2$ $equivalent^{-1}$ respecitively. **EC0258**

**45.** The resistance of a conductivity cell filled with 0.01N solution of NaCl is 200 ohm at18°C.Calculate the equivalent conductivity in S $cm^2$ $equivalent^{-1}$ of the solution. The cell constant of the conductivity cell is 0.88 $cm^{-1}$. **EC0259**

**46.** The molar conductivity of 0.1 M $CH_3COOH$ solution is 4 S $cm^2$ $mole^{-1}$ . What is the resistivity (in ohm-cm) of the solution ? **EC0260**

**47.** The conductivity of pure water in a conductivity cell with electrodes of cross sectional area 4 $cm^2$ and 2 cm apart is 8 × $10^{-7}$ S $cm^{-1}$. **EC0261**

(i) What is resistance (in kΩ) of conductivity cell ?

(ii) What current (in μA) would flow through the cell under an applied potential difference of 1 volt?

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**48.** 0.1M aq. $Na_2SO_4$ solution is filled in a conductivity cell having cell constant 0.2 $cm^{-1}$. Find the current in milli-amperes if 4 volt source is applied - **EC0262**

Given : $\Lambda_m(Na_2SO_4) = 2$ $ohm^{-1}cm^2mol^{-1}$ .

**49.** For 0.01N KCl, the resistivity 800 ohm cm. Find equivalent conductivity of KCl in S $cm^2$ $equivalent^{-1}$ ? **EC0263**

## APPLICATION OF KOHLRAUSCH'S LAW

**50.** $\Lambda_m^o$ for $CH_3COOH$ based on the data [All data in S–$cm^2$ $mol^{-1}$]. **EC0264**

| Electrolyte | $CaCl_2$ | $(CH_3COO)_2Ca$ | HCl |
|---|---|---|---|
| $\Lambda_m^o$ | 1400 | 800 | 700 |

**51.** For the strong electroytes NaOH, NaCl and $BaCl_2$ the molar ionic conductivities at infinite dilution are 240 $\times 10^{-4}$, 125 $\times 10^{-4}$ and 280.0 $\times 10^{-4}$ mho $m^2$ $mol^{-1}$ respectively. Calculate the molar conductivity in (mho $cm^2$ $mol^{-1}$) of $Ba(OH)_2$ at infinite dilution. **EC0265**

**52.** Molar conductance of $BaCl_2$, $H_2SO_4$ and HCl at infinite dilutions are 300 $\Omega^{-1}cm^2$/mole, 800 $\Omega^{-1}cm^2$/mole and 400 $\Omega^{-1}cm^2$/mole respectively. Equivalent conductance of $BaSO_4$ at infinite dilution is. (in S-$cm^2$/eq.) **EC0266**

**53.** Equivalent conductance of 0.01 N $Na_2SO_4$ solution is 120 $ohm^{-1}$ $cm^2$ $eq^{-1}$. The equivalent conductance at infinite dilution is 150 $ohm^{-1}$ $cm^2eq^{-1}$. What is the degree of dissociation in 0.01 N $Na_2SO_4$ solution?

**EC0205**

**54.** The value of $\Lambda_m^\infty$ for HCl, NaCl and $CH_3CO_2Na$ are 425, 125 and 100 S $cm^2$ $mol^{-1}$ respectively. If the equivalent conductivity of A given acetic acid is 48 S $cm^2$ $equivalent^{-1}$ at 25ºC, calculate it's degree of dissociation.

**EC0034**

**55.** Hydrofluoric acid is weak acid. At 25°C, the molar conductivity of 0.002M HF is 200 $ohm^{-1}$ $cm^2$ $mole^{-1}$. If its $\Lambda_m^\infty$ = 400 $ohm^{-1}$ $cm^2$ $mole^{-1}$, calculate its degree of dissociation and $pK_a$.

**EC0033**

**56.** Conductivity of an aqueous solution of 0.1 M HX (a weak mono-protic acid) is $5 \times 10^{-4}$ $Sm^{-1}$.

**Given :** $\Lambda_m^\infty[H^+] = 0.04$ $Sm^2mol^{-1}$

$\Lambda_m^\infty[X^-] = 0.01$ $Sm^2mol^{-1}$

Find $pK_a$ of HX. **EC0267**

**57.** At 25°C, $\lambda_\infty(H^+) = 3.5 \times 10^{-2}$ S $m^2$ $mol^{-1}$ and $\lambda_\infty(OH^-) = 2 \times 10^{-2}$ S $m^2mol^{-1}$. **EC0268**

Given: Specific conductance = $5.5 \times 10^{-6}$ S $m^{-1}$ for $H_2O$, determine pH & $pK_w$ for water.

(Give your answer as nearest integer).



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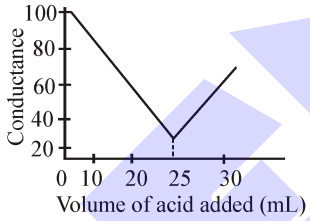
**58.** Saturated solution of AgCl at 25°C has specific conductance of $1.12\times10^{-6}$ ohm$^{-1}$ cm$^{-1}$. The $\lambda_{\infty}$ ($Ag^+$) and $\lambda_{\infty}$($Cl^-$) are 54 and 58 ohm$^{-1}$ cm$^2$ /equivalent respectively. Calculate $pK_{sp}$ of AgCl at 25°C.

**EC0269**

**59.** Specific conductance of a saturated solution of AgBr is $8.075\times10^{-7}$ ohm$^{-1}$cm$^{-1}$ at $25^0$C. Specific conductance of pure water at 25°C is $0.075\times10^{-7}$ ohm$^{-1}$ cm$^{-2}$. $\Lambda_m^{\infty}$ for KBr , $AgNO_3$ and $KNO_3$ are 140 , 130 , 110 ( S cm$^2$ mol$^{-1}$) respectively. If the solubility of AgBr is (A × $10^{-4}$ gm/litre) then value of A will be ? [Atomic mass : Ag = 108 , Br = 80]

**EC0032**

**60.** 20 mL of KOH solution was titrated with 0.20 mol L$^{-1}$ $H_2SO_4$ solution in a conductivity cell. The data obtained were plotted to give the graph shown below -




The concentration of the KOH solution (in mol/L) was. **EC0270**

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## EXERCISE # S-II

**1.** Calculate the emf of the cell in volts : [log (1.8) = 0.26]
Pt, $H_2$(1.0 atm) | $CH_3COOH$ (0.1M) || $NH_3$(aq, 0.01M) | $H_2$ (1.0 atm), Pt
$K_a(CH_3COOH) = 1.8 \times 10^{-5}$, $K_b$ $(NH_3) = 1.8 \times 10^{-5}$.
**EC0036**

**2.** The Edison storage cell is represented as Fe(s) | FeO(s) | KOH(aq) | $Ni_2O_3$ (s) | NiO(s) The half–cell reaction are
$Ni_2O_3(s) + H_2O(i) + 2e^- \rightleftharpoons 2NiO(s) + 2OH^-$, $E^0 = + 0.40V$
$FeO(s) + H_2O(l) + 2e^- \rightleftharpoons Fe(s) + 2OH^-$, $E^0 = - 0.87V$
(i) What is the cell e.m.f. in volts ?
(ii) What is the maximum amount of electrical energy (in kJ) that can be obtained from one mole of $Ni_2O_3$?
**EC0037**

**3.** The standard reduction potential for $Cu^{2+}$ | Cu is 0.34 V. Calculate the reduction potential in volts at pH = 14 for the above couple. $K_{sp}$ of $Cu(OH)_2$ is $1 \times 10^{-19}$.
**EC0038**

**4.** The emf of the cell Ag | AgI | KI (0.05M) || $AgNO_3$(0.05M) | Ag is 0.788V. If solubility product of AgI is 1. $10 \times 10^{-x}$ then value of x will be. $[10^{-0.788/0.059} = 4.4 \times 10^{-14}]$ **EC0271**

**5.** Consider the cell Ag | AgBr(s) $|Br^-||$ $Cl^-$ | AgCl(s) | Ag at 25° C . The solubility product constants of AgBr & AgCl are respectively $5 \times 10^{-13}$ & $1 \times 10^{-10}$ . For what ratio of the concentrations of $Cl^-$ & $Br^-$ ions would the emf of the cell be zero ?
**EC0208**

**6.** For the galvanic cell : Ag | AgCl(s) | KCl (0.2M) || K Br (0.001 M)| AgBr(s) | Ag, **EC0272**
Calculate the EMF of this cell in volts? [log (4.24) = 0.627]

$[K_{sp(AgCl)} = 2.8\times10^{-10}; K_{sp(AgBr)} = 3.3\times10^{-13}]$

**7.** Given, E° = –0.268 V for the $Cl^-$ | $PbCl_2$ | Pb couple and – 0.126 V for the $Pb^{2+}$ | Pb couple, If solubility product of $PbCl_2$ is $154 \times 10^{-x}$ then value of x will be at 25ºC.
$[10^{-0.284/0.059} = 1.54 \times 10^{-5}]$ **EC0273**

**8.** If equilibrium constant for reaction given below is $10^x$ then value of x will be. **EC0274**
$3Sn(s) + 2Cr_2O_7^{2-} + 28H^+ \longrightarrow 3Sn^{4+} + 4Cr^{3+} + 14H_2O$
$E^0$ for Sn | $Sn^{2+}$ = 0.136 V $E^0$ for $Sn^{2+}|Sn^{4+}$ = – 0.154 V
$E^0$ for $Cr_2O_7^{2-}|Cr^{3+}$ = 1.33 V

$\left[\frac{15.852}{0.059} = 268\right]$

(Give your answer as nearest integer).

**9.** One of the methods of preparation of per disulphuric acid, $H_2S_2O_8$, involve electrolytic oxidation of $H_2SO_4$ at anode ($2H_2SO_4 \rightarrow H_2S_2O_8 + 2H^+ + 2e^-$) with oxygen and hydrogen as by–products. In such an electrolysis, 2.27 L of $H_2$ and 0.5675 L of $O_2$ were generated at STP. What is the weight of $H_2S_2O_8$ formed (in grams)? **EC0275**

**10.** A current of 3 amp was passed for 2 hours through a solution of $CuSO_4$ ,3 g of $Cu^{2+}$ ions were deposited as Cu at cathode. Calculate percentage current efficiency of the process.
[Atomic mass : Cu = 63.5] **EC0041**

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**11.** Dal lake has water $8.2 \times 10^{12}$ litre approximately. A power reactor produces electricity at the rate of $1.5 \times 10^6$ coulomb per second at an appropriate voltage. How many milion years would it take to electrolyse the lake? [1 milion = $10^6$] **EC0276**

**12.** The equivalent conductance of 0.10 N solution of $MgCl_2$ is 97.1 mho $cm^2$ $equi^{-1}$ at 25°C. a cell with electrode that are 1.5 $cm^2$ in surface area and 0.5 cm apart is filled with 0.1 N $MgCl_2$ solution. How much current in amperes will flow when potential difference between the electrodes is 5 volt. **EC0277**

**13.** When a solution of specific conductance 1.342 $ohm^{-1}$ $metre^{-1}$ was placed in a conductivity cell with parallel electrodes, the resistance was found to be 170.5 ohm. Area of electrodes is $1.86\times10^{-4}$ $m^2$. Calculate separation of electrodes in metres.

**EC0043**

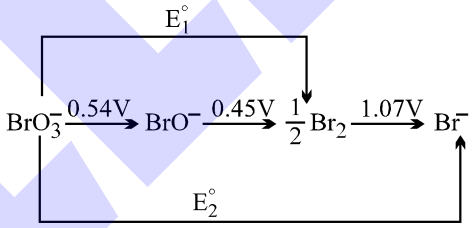
**14.** The specific conductance at 25°C of a saturated solution of $SrSO_4$ is $1.482\times10^{-4}$ $ohm^{-1}$ $cm^{-1}$while that of water used is $1.5\times10^{-6}$ mho $cm^{-1}$. Determine at 25°C the solubility in gm per litre of $SrSO_4$ in water. Molar ionic conductance of $Sr^{2+}$ and $SO_4^{2-}$ ions at infinite dilution are 59.46 and 79.8 $ohm^{-1}$ $cm^2$ $mole^{-1}$ respectively. **[ Sr = 87.6 , S = 32 , O = 16 ]**

$\left[\frac{1.467}{139.26} = 1.05\times10^{-2}\right]$

**EC0213**

**15.** The EMF of the cell $M \mid M^{n+}(0.02M) \parallel H^+(1M) \mid H_2(g)$ (1 atm), Pt at 25°C is 0.81V. Calculate the valency of the metal if the standard oxidation potential of the metal is 0.76V. **EC0278**

**16.** From the standard potentials shown in the following diagram, calculate the potentials $E_1^\circ$ and $E_2^\circ$ in volts.




**EC0215**

**17.** Calculate the EMF of the cell in volts,

$Zn - Hg(c_1M) \mid Zn^{2+}(aq) \mid Hg - Zn(c_2M)$

at 25°C, if the concentrations of the zinc amalgam are: $c_1$ = 10g per 100g of mercury and $c_2$ = 1g per100 g of mercury.

**EC0044**

**18.** For how much time in seconds a current of 2A has to be passed through a solution of $AgNO_3$ to coat a metal surface of $80cm^2$ with 5µm thick layer? Density of silver = 10.8g $cm^{-3}$. **EC0279**

**19.** 10g solution of $CuSO_4$ is electrolyzed using 0.01F of electricity. Calculate: **EC0280**

(a) The weight of resulting solution in gms

(b) Equivalents of acid or alkali in the solution.

**20.** Cadmium amalgam is prepared by electrolysis of a solution of $CdCl_2$ using a mercury cathode. How long in seconds should a current of 5A be passed in order to prepare 12% Cd – Hg amalgam on a cathode of 2 gm Hg (Cd = 112.4)

**EC0217**

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# EXERCISE # O-I

**NOTE :** If not given in question use : $\frac{2.303RT}{F} = 0.059$

## GALVANIC CELL

**1.** A standard hydrogen electrode has zero electrode potential because

(A) hydrogen is easier to oxidise (B) electrode potential is assumed to be zero

(C) hydrogen atom has only one electron (D) hydrogen is the lighest element.

**EC0047**

**2.** The thermodynamic efficiency of cell is given by-

(A) $\frac{\Delta H}{\Delta G}$ (B) $\frac{nFE_{cell}}{\Delta G}$ (C) $-\frac{nFE_{cell}}{\Delta H}$ (D) Zero

**EC0050**

**3.** From the following E° values of half cells,

(i) $A + e \rightarrow A^-$; E° = –0.24 V (ii) $B^- + e \rightarrow B^{2-}$; E° = +1.25 V

(iii) $C^- + 2e \rightarrow C^{3-}$; E° = –1.25 V (iv) $D + 2e \rightarrow D^{2-}$; E° = +0.68 V

What combination of two half cells would result in a cell with the largest potential ?

(A) (ii) and (iii) (B) (ii) and (iv) (C) (i) and (iii) (D) (i) and (iv)

**EC0055**

**4.** Which of the following will increase the voltage of the cell with following cell reaction

$Sn_{(s)} + 2Ag^+_{(aq)} \rightarrow Sn^{+2}_{(aq)} + 2Ag_{(s)}$

(A) Decrease in the concentration of $Ag^+$ ions (B) Increase in the concentration of $Sn^{+2}$ ions

(C) Increase in the concentration of $Ag^+$ ions (D) (A) & (B) both

**EC0059**

**5.** The standard electrode potentials for the reactions

$Ag^+ (a) + e^- \longrightarrow Ag(s)$ $Sn^{2+} (a) + 2e^- \longrightarrow Sn (s)$

at 25 °C are 0.80 volt and –0.14 volt, respectively. The standard emf of the cell.

$Sn_{(s)}|Sn^{2+}_{(aq)}(1M)||Ag^+_{(aq)}(1M)|Ag_{(s)}$ is :

(A) 0.66 volt (B) 0.80 volt (C) 1.08 volt (D) 0.94 volt

**EC0049**

**6.** $E°(Ni^{2+}|Ni)$ = –0.25 volt, $E° (Au^{3+} | Au)$ = 1.50 volt. The standard emf of the voltaic cell.

$Ni_{(s)} | Ni^{2+}_{(aq)} (1.0\ M) \| Au^{3+}_{(aq)} (1.0\ M) | Au_{(s)}$ is :

(A) 1.25 volt (B) –1.75 volt (C) 1.75 volt (D) 4.0 volt

**EC0053**

**7.** E° for $F_2 + 2e^- = 2F^-$ is 2.8 V, E° for $\frac{1}{2}F_2 + e^- = F^-$ is ?

(A) 2.8 V (B) 1.4 V (C) –2.8 V (D) –1.4 V

**EC0054**

**8.** If Δ G° of the cell reaction,

$AgCl(s) + \frac{1}{2}H_2(g) \rightarrow Ag(s) + H^+ + Cl^-$ is –21.52 KJ then

ΔG° of $2AgCl(s) + H_2(g) \rightarrow 2Ag(s) + 2H^+ + 2Cl^-$ is :

(A) –21.52 KJ (B) –10.76 KJ (C) –43.04 KJ (D) 43.04 KJ

**EC0056**

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**9.** The standard emf for the cell reaction,
$Zn_{(s)} + Cu^{2+}_{(aq)} \longrightarrow Zn^{2+}_{(aq)} + Cu_{(s)}$ is 1.10 volt at 25 °C. The emf for the cell reaction when 0.1 M $Cu^{2+}$ and 0.1 M $Zn^{2+}$ solution are used at 25°C is :
(A) 1.10 volt (B) 0.110 volt (C) –1.10 volt (D) –0.110 volt

**EC0064**

**10.** Given : $E^0_{Cr^{3+}/Cr} = -0.72$ V, $E^0_{Fe^{2+}/Fe} = -0.42$V. The potential for the cell
$Cr_{(s)} \mid Cr^{3+}_{(aq)}$ (0.1 M) $\| Fe^{2+}_{(aq)}$ (0.01 M)$|Fe_{(s)}$ is
(A) 0.26 V (B) 0.339 V (C) – 0.339 V (D) – 0.26 V

**EC0218**

**11.** What is the potential of the cell containing two hydrogen electrodes as represented below
$Pt \mid H_2(g) \mid H^+_{(aq)}(10^{-8}$ M) $\| H^+_{(aq)}(0.001$ M) $\mid H_2(g)|Pt$
(A) – 0.295 V (B) – 0.0591 V (C) 0.295 V (D) 0.0591 V

**EC0065**

**12.** Consider the cell, $Cu|Cu^{+2}||Ag^+|Ag$. If the concentration of $Cu^{+2}$ and $Ag^+$ ions becomes ten times the emf of the cell :–
(A) Becomes 10 times (B) Remains same
(C) Increase by 0.0295 V (D) Decrease by 0.0295 V

**EC0066**

**13.** The standard emf of a galvanic cell involving cell reaction with n = 4 is found to be 0.295 V at 25°C. The equilibrium constant of the reaction would be,
(A) $1.0 \times 10^{20}$ (B) $2.0 \times 10^{11}$ (C) $4.0 \times 10^{12}$ (D) $1.0 \times 10^{2}$

**EC0068**

**14.** The cell $Zn \mid Zn^{+2}_{(aq)}(1M) \| Cu^{+2}_{(aq)}{}^{(1M)} \mid Cu$ ($E^\circ_{cell} = 1.10$ V) was allowed to be completely discharged at 298 K. The relative concentration of $Zn^{+2}$ to $Cu^{+2}$, $\left\{\frac{[Zn^{+2}]}{[Cu^{+2}]}\right\}$ is :
(A) $9.65 \times 10^4$ (B) Antilog (24.08) (C) 37.3 (D) $10^{37.3}$

**EC0219**

**15.** Given the data at 25°C,
$Ag_{(s)} + I^-_{(aq)} \rightarrow AgI_{(s)} + e^-$, $E^\circ = 0.152$V
$Ag_{(s)} \rightarrow Ag^+_{(aq)} + e^-$, $E^\circ = -0.800$ V
What is the value of log $K_{sp}$ for AgI ? (Where $K_{sp}$= solubility product)
$\left(2.303\frac{RT}{F} = 0.059V\right)$
(A) –8.12 (B) +8.612 (C) –37.83 (D) –16.13

**EC0220**

**16.** In a cell that utilises the reaction,
$Zn(s) + 2H^+(A) \longrightarrow Zn^{2+}(aq) + H_2(g)$ addition of $H_2SO_4$ to cathode compartment, will :
(A) increase the $E_{cell}$ and shift equilibrium to the right
(B) lower the $E_{cell}$ and shift equilibrium to the right
(C) lower the $E_{cell}$ and shift equilibrium to the left
(D) increase the $E_{cell}$ and shift equilibrium to the left

**EC0221**

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**17.** By how much times will potential of half cell $Cu^{+2}|Cu$ change if, the solution is diluted to 100 times at 298 K :–

(A) Increases by 59 mV (B) Decrease by 59 mV

(C) Increases by 29.5 mV (D) Decreases by 29.5 mV

**EC0071**

**18.** The standard reduction electrode potentials of four metal cations are - **EC0281**

$E^0_{A^+/A} = -0.250$ V , $E^0_{B^+/B} = -0.140$ V $E^0_{C^+/C} = -0.126$ V , $E^0_{D^+/D} = -0.402$ V

The metal that displaces $A^+$ from its aqueous solution is :–

(A) B (B) C (C) D (D) None of the above

**19.** The reduction potential values are given below: **EC0282**

$E^0_{Al^{3+}/Al} = -1.67V$ ; $E^0_{Mg^{2+}/Mg} = -2.34V$ , $E^0_{Cu^{2+}/Cu} = 0.34V$; $E^0_{I_2/I^-} = 0.53V$

Which one is the best reducing agent ?

(A) Al (B) Mg (C) Cu (D) $I_2$

## ELECTROLYTIC CELL

**20.** When an electric current is passed through a cell containing an electrolyte, positive ions move towards the cathode and negative ions towards the anode. What will happen if the cathode is pulled out of the solution?

(A) The positive ions will start moving towards the anode and negative ions will stop moving.

(B) The negative ions will continue to move towards the anode and the positive ions will stop moving

(C) Both positive and negative ions will move towards the anode.

(D) None of these movements will take place.

**EC0072**

**21.** The products formed when an aqueous solution of NaBr is electrolyzed in a cell having inert electrodes are :

(A) Na and $Br_2$ (B) Na and $O_2$

(C) $H_2$, $Br_2$ and NaOH (D) $H_2$ and $O_2$

**EC0074**

**22.** A solution of sodium sulphate in water is electrolysed using inert electrodes. The products at the cathode and anode are respectively.

(A) $H_2$, $O_2$ (B) $O_2$, $H_2$ (C) $O_2$, Na (D) none

**EC0076**

**23.** When an aqueous solution of lithium chloride is electrolysed using graphite electrodes

(A) $Cl_2$ is liberated at the anode.

(B) Li is deposited at the cathode

(C) as the current flows, pH of the solution remains constant

(D) as the current flows, pH of the solution decreases.

**EC0077**

**24.** The Gibbs energy for the decomposition of $Al_2O_3$ at 500 °C is as follows

$$\frac{2}{3}Al_2O_3 \longrightarrow \frac{4}{3}Al + O_2,\ \Delta_rG = +966\ KJ\ mol^{-1}$$

The potential difference needed for electrolytic reduction of $Al_2O_3$ at 500 °C is at least :-

(A) 5.0 V (B) 4.5 V (C) 3.0 V (D) 2.5 V

**EC0222**

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**25.** Electrolysis of a $CuSO_4$ produces :–
(A) An increase in pH (B) A decrease in pH
(C) Either decrease or increase (D) None
**EC0075**

**26.** The amount of an ion discharged during electrolysis is not directly proportional to :
(A) resistance
(B) time
(C) current strength
(D) electrochemical equivalent of the element
**EC0078**

**27.** Number of electrons involved in the electrodeposition of 63.5 g of Cu from a solution of $CuSO_4$ is : **($N_A = 6 \times 10^{23}$)**
(A) $6 \times 10^{23}$ (B) $3 \times 10^{23}$ (C) $12 \times 10^{23}$ (D) $6 \times 10^{22}$
**EC0079**

**28.** When one coulomb of electricity is passed through an electrolytic solution the mass deposited on the electrode is equal to :
(A) equivalent weight (B) molecular weight
(C) electrochemical equivalent (D) one gram
**EC0080**

**29.** Electro chemical equivalent of a substance is 0.0006; its equivalent weight. is :
(A) 57.9 (B) 28.95
(C) 115.8 (D) cannot be calculated
**EC0081**

**30.** The same amount of electricity was passed through two separate electrolytic cells containing solutions of nickel nitrate $[Ni(NO_3)_2]$and chromium nitrate $[Cr(NO_3)_3]$ respectively. If 0.3 g of nickel was deposited in the first cell, the amount of chromium deposited is : **(at. wt. of Ni = 59, at. wt. of Cr = 52)**
(A) 0.1 g (B) 0.17 g (C) 0.3 g (D) 0.6 g
**EC0087**

**31.** W g of copper deposited in a copper voltameter when an electric current of 2 ampere is passed for 2 hours. If one ampere of electric current is passed for 4 hours in the same voltameter, copper doposited will be :
(A) W (B) W|2 (C) W|4 (D) 2W
**EC0082**

**32.** When the same electric current is passed through the solution of different electrolytes in series the amounts of elements deposited on the electrodes are in the ratio of their:
(A) atomic number (B) atomic masses (C) specific gravities (D) equivalent masses
**EC0083**

**33.** The amount of electricity that can deposit 108 g. of silver from silver nitrate solution is:
(A) 1 ampere (B) 1 coulomb
(C) 1 Faraday (D) 2 ampere
**EC0084**

**34.** The ratio of weights of hydrogen and magnesium deposited by the same amount of electricity from aqueous $H_2SO_4$ and fused $MgSO_4$ are :
(A) 1 : 8 (B) 1 : 12 (C) 1 : 16 (D) None of these
**EC0085**

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**35.** A current of 9.65 amp. flowing for 10 minute deposits 3.0 g of a metal. The equivalent wt. of the metal is :
(A) 10 (B) 30 (C) 50 (D) 96.5

**EC0086**

**36.** 1 mole of Al is deposited by X coulomb of electricity passing through aluminium nitrate solution. The number of moles of silver deposited by X coulomb of electricity from silver nitrate solution is
(A) 3 (B) 4 (C) 2 (D) 1

**EC0092**

**37.** The electric charge for electro deposition of 1 equivalent of a substance is :
(A) one ampere per second (B) 4 faraday
(C) one ampere for one hour (D) charge on one mole of electrons

**EC0088**

**38.** 3.17 g., of a substance was deposited by the flow of 0.1 mole of electrons. The equivalent weight of the substance is :
(A) 3.17 (B) 0.317 (C) 317 (D) 31.7

**EC0089**

**39.** A current of 9.65 amp. passing for 16 min. 40 sec. through a molten tin salt deposits 5.95 g. of tin The oxidation state of the tin in the salt is : (at. wt of Sn = 119)
(A) +4 (B) +3 (C) +2 (D) +1

**EC0090**

**40.** The time required for a current of 3 amp. to decompose electrolytically 18 g of $H_2O$ is:
(A) 18 hour (B) 36 hour (C) 9 hour (D) 18 seconds

**EC0091**

**41.** An ion is reduced to the element when it absorbs $6 \times 10^{20}$ electrons. The number of gm equivalents of the ion is :
(A) 0.10 (B) 0.01 (C) 0.001 (D) 0.0001

**EC0093**

**42.** How many coulombs of electric charge are required for the oxidation of 1 mole of $H_2O$ to $O_2$ ?
(A) $9.65 \times 10^4$ C (B) $4.825 \times 10^5$ C
(C) $1.93 \times 10^5$ C (D) $1.93 \times 10^4$ C

**EC0095**

**43.** The time required to coat ameter surface of 80 $cm^2$ with $5 \times 10^{-3}$ cm thick layer of silver (density 1.08 g $cm^{-3}$) with the passage of 9.65A current through a silver nitrate solution is :
(A) 10 sec. (B) 40 sec. (C) 30 sec. (D) 20 sec.

**EC0099**

**44.** One gm metal $M^{+2}$ was discharged by the passage of $1.2 \times 10^{22}$ electrons. What is the atomic weight of metal?
(A) 25 (B) 50 (C) 100 (D) 75

**EC0100**

**45.** One mole of electron passes through each of the solution of $AgNO_3$, $CuSO_4$ and $AlCl_3$ when Ag, Cu and Al are deposited at cathode. The molar ratio of Ag, Cu and Al deposited are
(A) 1 : 1 : 1 (B) 6 : 3 : 2 (C) 6 : 3 : 1 (D) 1 : 3 : 6

**EC0101**

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**46.** During electrolysis of an aqueous solution of sodium sulphate, 2.4 L of oxygen at STP was liberated at anode. The volume of hydrogen at STP, liberated at cathode would be

(A) 1.2 L (B) 2.4 L (C) 2.6 L (D) 4.8 L

**EC0102**

**47.** The charge required for the oxidation of one mole $Mn_3O_4$ into $MnO_4^{2-}$ in presence of alkaline medium is

(A) $5 \times 96500$ C (B) 96500 C (C) $10 \times 96500$ C (D) $2 \times 96500$ C

**EC0103**

## *CONDUCTANCE*

**48.** Resistance of a conductivity cell filled with a solution of an electrolyte of concentration 0.1M is 100Ω. The conductivity of this solution is 1.29 $Sm^{-1}$. Resistance of the same cell when filled with 0.02M of the same solution is 520Ω. The molar conductivity of 0.02M solution of the electrolyte will be.

(A) $124 \times 10^{-4}$ $Sm^2$ $mol^{-1}$ (B) $1240 \times 10^{-4}$ $Sm^2$ $mol^{-1}$

(C) $1.24 \times 10^4$ $Sm^2$ $mol^{-1}$ (D) $12.4 \times 10^{-4}$ $Sm^2$ $mol^{-1}$

**EC0227**

**49.** If x is specific resistance of the electrolyte solution and y is the molarity of the solution, then $\Lambda_m$ is given by

(A) $\frac{1000\,x}{y}$ (B) $1000\,\frac{y}{x}$ (C) $\frac{1000}{xy}$ (D) $\frac{xy}{1000}$

**EC0115**

**50.** If 0.01 M solution of an electrolyte has a resistance of 40 ohms in a cell having a cell constant of 0.4 $cm^{-1}$ then its molar conductance in $ohm^{-1}$ $cm^2$ $mol^{-1}$ is : **EC0283**

(A) 10 (B) $10^2$ (C) $10^3$ (D) $10^4$

**51.** The specific conductance of a 0.01 M solution of KCl is 0.0014 $ohm^{-1}$ $cm^{-1}$at 25° C. Its equivalent conductance ($cm^2$ $ohm^{-1}$ $equiv^{-1}$) is :– **EC0284**

(A) 140 (B) 14 (C) 1.4 (D) 0.14

**52.** The resistance of 0.1 N solution of a acetic acid is 250 ohm. When measured in a cell of cell constant 1.15 $cm^{-1}$. The equivalent conductance (in $ohm^{-1}$ $cm^2$ equiv.$^{-1}$) of 0.1 N acetic acid is: **EC0285**

(A) 46 (B) 9.2 (C) 18.4 (D) 0.023

**53.** Electrolytic conduction differs from metallic conduction from the fact that in the former **EC0286**

(A) The resistance increases with increasing temperature

(B) The resistance decreases with increasing temperature

(C) The resistance remains constant with increasing temperature

(D) The resistance is independent of the length of the conductor

**54.** The highest electrical conductivity of the following aqueous solution is of

(A) 0.1 M fluoroacetic acid (B) 0.1 M difluoroacetic acid

(C) 0.1 M acetic acid (D) 0.1 M chloroacetic acid

**EC0225**

**55.** Which of the following solution of KCl has the lowest value of specific conductance :

(A) 1 M (B) 0.1 M (C) 0.01 M (D) 0.001 M **EC0287**

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**56.** Which of the following solutions of KCl has the lowest value of equivalent conductance ?
(A) 1 M (B) 0.1 M (C) .01 M (D) .001 M **EC0288**

**57.** The limiting molar conductivities $\Lambda^0$ for NaCl, KBr and KCl are 126,152 and 150 $Scm^2\ mol^{-1}$ respectively. The $\Lambda_m^0$ for NaBr is :
(A) 278 S $cm^2\ mol^{-1}$ (B) 176 S $cm^2\ mol^{-1}$
(C) 128 S $cm^2\ mol^{-1}$ (D) 302 S $cm^2\ mol^{-1}$
**EC0223**

**58.**

| **Electrolyte** | **$\Lambda^\infty$ (S $cm^2\ mol^{-1}$)** |
|---|---|
| KCl | 149.9 |
| $KNO_3$ | 145.0 |
| HCl | 426.2 |
| NaOAC | 91.0 |
| NaCl | 126.5 |

Calculate $\Lambda^\infty_{HOAc}$ using appropriate molar conductances of the electrolytes listed above at infinite dilution in $H_2O$ at 25°C
(A) 390.7 (B) 217.5 (C) 517.2 (D) 552.7
**EC0224**

**59.** The molar conductivities, $\Lambda^0_{NaOAc}$ and $\Lambda^0_{HCl}$ at infinite dilution in water at 25°C are 91.0 and 426.2 S $cm^2$|mol respectively. To calculate $\Lambda^0_{HOAc}$ the additional value required is :
(A) KCl (B) NaOH (C) NaCl (D) $H_2O$
**EC0226**

**60.** The equivalent conductivity of 0.1 N $CH_3COOH$ at 25 °C is 80 S-$cm^2$/eq. and at infinite dilution 400 S-$cm^2$/eq. The degree of dissociation of $CH_3COOH$ in the solution is :
(A) 1 (B) 0.2 (C) 0.1 (D) 0.5 **EC0289**

**61.** Equivalent conductances of $Ba^{+2}$ and $Cl^-$ ions are 127 & 76 $ohm^{-1}\ cm^2\ eq^{-1}$ respectively. Equivalent conductance of $BaCl_2$ at infinite dilution is -
(A) 379 (B) 139.5 (C) 203 (D) 330
**EC0114**

**62.** The conductivity of a saturated solution of $BaSO_4$ is $3.06 \times 10^{-6}$ $ohm^{-1}\ cm^{-1}$ and its molar conductance is 1.53 $ohm^{-1}\ cm^2\ mol^{-1}$. The $K_{sp}$ of $BaSO_4$ will be :
(A) $2 \times 10^{-4}$ (B) $4 \times 10^{-4}$ (C) $4 \times 10^{-3}$ (D) $4 \times 10^{-6}$
**EC0113**

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## EXERCISE # O-II

**Single correct :**

**1.** Consider the reaction, $Cl_{2(g)} + 2Br^-_{(aq)} \longrightarrow 2Cl^-_{(aq)} + Br_{2(g)}$
The emf of the cell when $[Cl^-]=[Br^-]=0.01M$ and $Cl_2$ gas at 1 atm pressure while $Br_2(g)$ at 0.01 atm will be ($E^\circ$ for the above reaction is = 0.29 volt) :
(A) 0.54 volt (B) 0.35 volt (C) 0.24 volt (D) –0.29 volt
**EC0116**

**2.** How much will the reduction potential of a hydrogen electrode change when its solution initially at pH = 0 is neutralised to pH = 7 ?
(A) increase by 0.059V (B) decrease by 0.059V
(C) increase by 0.413V (D) decrease by 0.413V
**EC0117**

**3.** If the pressure of $H_2$ gas is increased from 1 atm to 100 atm keeping $H^+$ concentration constant at 1 M, the change in reduction potential of hydrogen half cell at 25°C will be
(A) 0.059 V (B) 0.59 V (C) 0.0295 V (D) 0.118 V
**EC0228**

**4.** A silver wire dipped in 0.1 M HCl solution saturated with AgCl develops oxidation potential of –0.209 V. If $E^\circ_{Ag/Ag^+} = -0.799$ V, the $K_{sp}$ of AgCl in pure water will be
(A) $3 \times 10^{-11}$ (B) $10^{-11}$ (C) $4 \times 10^{-11}$ (D) $3 \times 10^{-11}$
**EC0229**

**5.** Salts of A (atomic weight = 7), B (atomic weight =27) and C (atomic weight = 48) were electrolysed under identical conditions using the same quantity of electricity. It was found that when 2.1 g of A was deposited, the weights of B and C deposited were 2.7 and 7.2 g. The valencies of A, B and C respectively are
(A) 3, 1 and 2 (B) 1, 3 and 2 (C) 3, 1 and 3 (D) 2, 3 and 2
**EC0230**

**6.** During electro refining of Cu by electrolysis of an aqueous solution of $CuSO_4$ using copper electrodes, if 2.5 g of Cu is deposited at cathode, then at anode
(A) decrease of more than 2.5 g of mass takes place (B) 450 ml of $O_2$ at STP is liberated
(C) 2.5 g of copper is deposited (D) a decrease of 2.5 g of mass takes place
**EC0118**

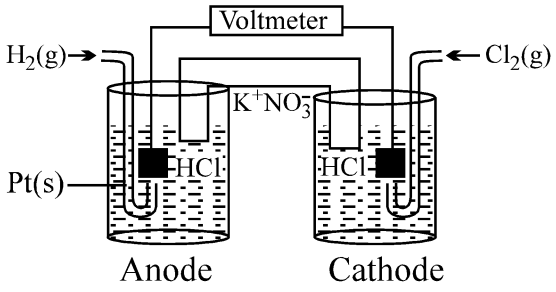
**7.** The conductivity of a saturated solution of $Ag_3PO_4$ is $9 \times 10^{-6}$ S $m^{-1}$ and its equivalent conductivity is $1.50 \times 10^{-4}$ S $m^2$ equivalent$^{-1}$. The $K_{sp}$ of $Ag_3PO_4$ is
(A) $4.32 \times 10^{-18}$ (B) $1.8 \times 10^{-9}$ (C) $8.64 \times 10^{-13}$ (D) None of these
**EC0119**

**8.** Equal volumes of 0.015 M $CH_3COOH$ & 0.015 M NaOH are mixed together. What would be molar conductivity of mixture if conductivity of $CH_3COONa$ is $6.3 \times 10^{-4}$ S $cm^{-1}$
(A) 8.4 S $cm^2$ $mol^{-1}$ (B) 84 S $cm^2$ $mol^{-1}$ (C) 4.2 S $cm^2$ $mol^{-1}$ (D) 42 S $cm^2$ $mol^{-1}$
**EC0120**

**9.** For the fuel cell reaction $2H_2(g) + O_2(g) \longrightarrow 2H_2O(l)$ ; $\Delta_f H^\circ_{298}(H_2O, l) = -285.5$ kJ|mol
What is $\Delta S^\circ_{298}$ for the given fuel cell reaction?
Given: $O_2(g) + 4H^+(aq) + 4e^- \longrightarrow 2H_2O(l)$ ; $E^\circ = 1.23$ V
(A) – 0.322 J|K (B) – 0.635 kJ|K (C) 3.51 kJ|K (D) – 0.322 kJ|K
**EC0231**

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**10.** Consider the following Galvanic cell.




By what value the cell voltage change when concentration of ions in anodic and cathodic compartments both increased by factor of 10 at 298 K

(A) +0.0590 (B) –0.0590 (C) –0.1180 (D) 0

**EC0121**

**11.** The standard reduction potentials at 25°C for the following half reactions are :

$Zn^{2+}(aq) + 2e^- \rightleftharpoons Zn(s)$, $E^\circ_{RP} = -0.762V$

$Cr^{3+}(aq) + 3e^- \rightleftharpoons Cr(s)$, $E^\circ_{RP} = -0.740V$

$2H^+_{(aq)} + 2e^- \rightleftharpoons H_2(g)$, $E^\circ_{RP} = 0.00\ V$

$Fe^{3+}_{(aq)} + 2e^- \rightleftharpoons Fe^{2+}_{(aq)}$, $E^\circ_{RP} = 0.77V$

Which is the strongest reducing agent ?

(A) Zn (B) Cr (C) $H_2(g)$ (D) $Fe^{2+}(aq)$

**EC0122**

**12.** Using the standard electrode potential values given below, decide which of the statements, I, II, III and IV are correct. Choose the right answer from (A), (B), (C) and (D).

$Fe^{2+}_{(aq)} + 2e^- \rightleftharpoons Fe_{(s)}$ ; $E^\circ = -0.44\ V$

$Cu^{2+}_{(aq)} + 2e^- \rightleftharpoons Cu_{(s)}$ ; $E^\circ = +0.34\ V$

$Ag^+_{(aq)} + e^- \rightleftharpoons Ag_{(s)}$ ; $E^\circ = +0.80\ V$

I. Copper can displace iron from $FeSO_4$ solution.

II. Iron can displace copper from $CuSO_4$ solution.

III. Silver can displace copper from $CuSO_4$ solution.

IV. Iron can displace silver from $AgNO_3$ solution.

(A) I and II (B) II and III (C) II and IV (D) I and IV

**EC0232**

**13.** The following facts are available :–

$2X^- + Y_2 \rightarrow 2Y^- + X_2$

$2W^- + Y_2 \rightarrow$ NO reaction

$2Z^- + X_2 \rightarrow 2X^- + Z_2$

Which of the following statements is correct :–

(A) $E^\circ_{W^-/W_2} > E^\circ_{Y^-/Y_2} > E^\circ_{X^-/X_2} > E^\circ_{Z^-/Z_2}$

(B) $E^\circ_{W^-/W_2} < E^\circ_{Y^-/Y_2} < E^\circ_{X^-/X_2} < E^\circ_{Z^-/Z_2}$

(C) $E^\circ_{W^-/W_2} < E^\circ_{Y^-/Y_2} > E^\circ_{X^-/X_2} > E^\circ_{Z^-/Z_2}$

(D) $E^\circ_{W^-/W_2} > E^\circ_{Y^-/Y_2} < E^\circ_{X^-/X_2} < E^\circ_{Z^-/Z_2}$

**EC0233**

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**14.** The cost of electricity required to deposit 1 g of Mg is Rs. 5.00. How much would it cost to deposit 9 g of Al (At wt. Al = 27, Mg = 24)

(A) Rs. 10 (B) Rs. 27 (C) Rs. 40 (D) Rs. 60

**EC0234**

**15.** 4.5g of aluminium (at. mass 27 amu) is deposited at cathode from $Al^{3+}$ solution by a certain quantity of electric charge. The volume of hydrogen produced at STP from $H^+$ ions in solution by the same quantity of electric charge will be –

(A) 45.4L (B) 11.35L (C) 22.7L (D) 5.675 L

**EC0123**

**16.** The density of A is 10 g $cm^{-3}$. The quantity of electricity needed to plate an area 10 cm × 10 cm to a thickness of $10^{-2}$ cm using $ASO_4$ solution would be (Atomic mass of A = 193)

(A) 5000 C (B) 10000 C (C) 40000 C (D) 20000 C

**EC0124**

**17.** The resistance of 0.5 M solution of an electrolyte in a cell was found to be 50 Ω. If the electrodes in the cell are 2.2 cm apart and have an area of 4.4 $cm^2$ then the molar conductivity (in S $m^2$ $mol^{-1}$) of the solution is

(A) 0.2 (B) 0.02 (C) 0.002 (D) None of these

**EC0125**

**18.** Equivalent conductance of 0.1 M HA(weak acid) solution is 10 $Scm^2equivalent^{-1}$ and that at infinite dilution is 200 $Scm^2equivalent^{-1}$ Hence pH of HA solution is

(A) 1.3 (B) 1.7 (C) 2.3 (D) 3.7

**EC0126**

**19.** The dissociation constant of n-butyric acid is $1.6 \times 10^{-5}$ and the molar conductivity at infinite dilution is $380 \times 10^{-4}$ $Sm^2mol^{-1}$. The specific conductance of the 0.01 M acid solution is

(A) $1.52 \times 10^{-5}$ $Sm^{-1}$ (B) $1.52 \times 10^{-2}$ $Sm^{-1}$

(C) $1.52 \times 10^{-3}$ $Sm^{-1}$ (D) None

**EC0127**

**Multiple correct :**

**20.** During discharging of lead storage battery, which of the following is|are true ?

(A) $H_2SO_4$ is produced (B) $H_2O$ is consumed

(C) $PbSO_4$ is formed at both electrodes (D) Density of electrolytic solution decreases

**EC0128**

**21.** Which of the following arrangement will produce oxygen at anode during electrolysis ?

(A) Dilute $H_2SO_4$ solution with Cu electrodes.

(B) Dilute $H_2SO_4$ solution with inert electrodes.

(C) Fused NaOH with inert electrodes.

(D) Dilute NaCl solution with inert electrodes.

**EC0235**

**22.** If 270.0 g of water is electrolysed during an experiment performed by miss Abhilasha with 75% current efficiency then

(A) 168 L of $O_2$ (g) will be evolved at anode at 1 atm & 273 K

(B) Total 504 L gases will be produced at 1 atm & 273 K.

(C) 336 L of $H_2$ (g) will be evolved at anode at 1 atm & 273 K

(D) 45 F electricity will be consumed

**EC0129**

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**23.** Pick out the correct statements among the following from inspection of standard reduction potentials (Assume standard state conditions).

$Cl_2$ (a) + 2e $\rightleftharpoons$ $2Cl^-$(a) $E^o_{Cl_2/Cl^-} = +1.36$ volt

$Br_2$ (a) + 2e $\rightleftharpoons$ $2Br^-$(a) $E^o_{Br_2/Br^-} = +1.09$ volt

$I_2$(s) + 2e $\rightleftharpoons$ $2I^-$ (a) $E^o_{I_2/I^-} = +0.54$ volt

$S_2O_8^{2-}$ (a) + 2e $\rightleftharpoons$ $2SO_4^{2-}$ (a) $E^o_{S_2O_8^{2-}/SO_4^{2-}} = +2.00$ volt

(A) $Cl_2$ can oxidise $SO_4^{2-}$ from solution

(B) $Cl_2$ can oxidise $Br^-$ and $I^-$ from aqueous solution

(C) $S_2O_8^{2-}$ can oxidise $Cl^-$, $Br^-$ and $I^-$ from aqueous solution

(D) $S_2O_8^{2-}$ is added slowly, $Br^-$ can be reduce in presence of $Cl^-$

**EC0236**

**24.** The EMF of the following cell is 0.22 volt.

Ag(s) | AgCl(s) | KCl (1M) | $H^+$(1M) | $H_2$(g) (1atm) ; Pt(s).

Which of the following will decrease the EMF of cell.

(A) increasing pressure of $H_2$(g) from 1 atm to 2 atm

(B) increasing $Cl^-$ concentration in Anodic compartment

(C) increasing $H^+$ concentration in cathodic compartment

(D) decreasing KCl concentration in Anodic compartment.

**EC0130**

**Assertion & Reasoning type questions :**

**25.** Statement -1 : The voltage of mercury cell remains constant for long period of time.

Statement -2 : It is because net cell reaction does not involve active species.

(A) Statement-1 is true, statement-2 is true and statement-2 is correct explanation for statement-1.

(B) Statement-1 is true, statement-2 is true and statement-2 is NOT the correct explanation for statement-1.

(C) Statement-1 is true, statement-2 is false.

(D) Statement-1 is false, statement-2 is true.

**EC0237**

**26.** Statement -1 : The SRP of three metallic ions $A^+$,$B^{2+}$, $C^{3+}$ are –0.3, –0.5, 0.8 volt respectively, so oxidising power of ions is $C^{3+} > A^+ > B^{2+}$.

Statement -2 : Higher the SRP, higher the oxidising power.

(A) Statement-1 is true, statement-2 is true and statement-2 is correct explanation for statement-1.

(B) Statement-1 is true, statement-2 is true and statement-2 is NOT the correct explanation for statement-1.

(C) Statement-1 is true, statement-2 is false.

(D) Statement-1 is false, statement-2 is true.

**EC0131**

**27.** Statement -1 : We can add the electrode potential in order to get electrode potential of net reaction.

Statement -2 : Electrode potential is an intensive property.

(A) Statement-1 is true, statement-2 is true and statement-2 is correct explanation for statement-1.

(B) Statement-1 is true, statement-2 is true and statement-2 is NOT the correct explanation for statement-1.

(C) Statement-1 is true, statement-2 is false.

(D) Statement-1 is false, statement-2 is true.

**EC0132**

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**Table type :**

**1 TABLE (3Q)**

| | Column-I | | Column-II | | Column-III |
|---|---|---|---|---|---|
| (P) | $Zn(s)\|ZnSO_4(0.1M)\|\|Zn(NO_3)(0.01M)Zn(s)$ | (A) | Has metal -insoluble salt - anion electrode. | (I) | $E_{cell} < 0$ |
| (Q) | $Pt,Cl_2(0.1bar)\|KCl(1M)\|\|NaCl(1M)\ Cl_2(1bar),Pt$ | (B) | Electrolytic concentration cell | (II) | $E^0_{cell} = 0$ |
| (R) | $Ag(s)\|\ AgCl(s)\|KCl(0.1M)\|\|Ag^+(0.1M)\|Ag(s)$ $Ksp[AgCl] = 10^{-10}$. | (C) | Electrode concentration cell | (III) | $E_{cell} > 0$ |
| (S) | $Pt,H_2(1bar)\|\ H_2SO_4(0.05M)\|\|HNO_3(0.1M)\|H_2$ (1bar), Pt | (D) | Has gas-ion electrode | (IV) | $E_{cell} = 0$ |

(1) Use : $\frac{2.303RT}{F} = 0.06$

(2) Assume constant P,T condition of operation.

**28.** Which option is incorrectly matched ?

(A) P - B - II (B) Q - C - II

(C) R - A - I (D) S - D - IV

**EC0133**

**29.** For galvanic cell in option 'Q' on increasing concentration of KCl, cell potential will -

(A) Increase (B) decrease

(C) remains constant (D) cannot predict

**EC0134**

**30.** On increasing $Ag^+$ concentration in anodic compartment in option (R) cell potential will

(A) Remain same (B) increase

(C) decrease (D) can't predict

**EC0135**

**Match the column**

**31.**

| | Column I | | Column II (Electrolysis product using inert electrode) |
|---|---|---|---|
| (A) | Dilute solution of HCl | (P) | $O_2$ evolved at anode |
| (B) | Dilute solution of NaCl | (Q) | $H_2$ evolved at cathode |
| (C) | Concentrated solution of NaCl | (R) | $Cl_2$ evolved at anode |
| (D) | $AgNO_3$ solution | (S) | Ag deposition at cathode |

**EC0136**

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**32.** **Column-I** **Column-II**

**Cell notation :** $E_{cell}$

(P) $Mn\left|{}^{Mn^{2+}}_{(0.1M)}\right\|{}^{Cu^{2+}}_{(0.01M)}\left|Cu\right.$ (1) 0.33

(Q) $Ag\left|AgBr\right|{}^{Br^-}_{(0.1M)}\left\|{}^{Ag^+}_{(0.01M)}\right|Ag$ (2) 0.6

(R) $Pt,H_2(4atm)\left|{}^{H^+}_{(0.01M)}\right\|{}^{Cl^-}_{(0.2M)}\left|{}^{Cl_2}_{(1atm)}\right|Pt$ (3) 1.49

(S) $Pt,\left|{}^{Fe^{2+}}_{(0.2M)}\ {}^{Fe^{+3}}_{(0.02M)}\right\|{}^{MnO_4^-}_{(0.1M)}\ {}^{Mn^{+2}}_{(0.1M)}\ {}^{H^+}_{pH=5}\left|Pt\right.$ (4) 1.54

Given :

$E^0_{Cu^{2+}/Cu} = 0.34V$ $K_{sp}(AgBr) = 10^{-13}$

$E^0_{Mn^{2+}/Mn} = -1.18V$ $\frac{2.303RT}{F} = 0.06$

$E^0_{Ag^+/Ag} = 0.8V$

$E^0_{Fe^{3+}/Fe^{2+}} = 0.77V$

$E^0_{MnO_4^-/Mn^{2+}} = 1.52V$

$E^0_{Cl_2/Cl^-} = 1.36V$

**Code :**

| | P | Q | R | S |
|---|---|---|---|---|
| (A) | 4 | 3 | 4 | 4 |
| (B) | 2 | 3 | 1 | 4 |
| (C) | 3 | 2 | 4 | 1 |
| (D) | 4 | 3 | 2 | 1 |

**EC0238**

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## EXERCISE # J-MAIN

**1.** Given **[JEE-MAINS - 2017]**

$E^{o}_{Cl_2/Cl^-} = 1.36\,V, E^{o}_{Cr^{3+}/Cr} = -0.74\,V$ ; $E^{o}_{Cr_2O_7^{2-}/Cr^{3+}} = 1.33\,V, E^{o}_{MnO_4^-/Mn^{2+}} = 1.51\,V$.

Among the following, the strongest reducing agent is

(1) Cr (2) $Mn^{2+}$ (3) $Cr^{3+}$ (4) $Cl^-$

**EC0148**

**2.** What is the standard reduction potential (E°) for $Fe^{3+} \rightarrow Fe$ ? **[JEE-MAINS (ONLINE) 2017]**

Given that :

$Fe^{2+} + 2e^- \rightarrow Fe$ ; $E^{o}_{Fe^{2+}/Fe} = -0.47$ V

$Fe^{3+} + e^- \rightarrow Fe^{2+}$ ; $E^{o}_{Fe^{3+}/Fe^{2+}} = +0.77$ V

(1) +0.30 V (2) +0.057 V (3) –0.057 V (4) –0.30 V

**EC0149**

**3.** To find the standard potential of $M^{3+}|M$ electrode, the following cell is constituted: $Pt|M|M^{3+}(0.001\ mol\ L^{-1})|Ag^+(0.01\ mol\ L^{-1})|Ag$ **[JEE-MAINS (ONLINE) 2017]**

The emf of the cell is found to be 0.421 volt at 298 K. The standard potential of half reaction $M^{3+} + 3e^- \rightarrow M$ at 298 K will be : **(Given $E^{\ominus}_{Ag^+/Ag}$ at 298 K = 0.80 Volt)**

(1) +0.30 V (2) +0.057 V (3) –0.057 V (4) –0.30 V

**EC0150**

**4.** How long (approximate) should water be electrolysed by passing through 100 amperes current so that the oxygen released can completely burn 27.66 g of diborane ?**[JEE-MAINS (OFFLINE) 2017]**

**(Atomic weight of B = 10.8 u)**

(1) 0.8 hours (2) 3.2 hours (3) 1.6 hours (4) 6.4 hours

**EC0151**

**5.** When an electric current is passed through acidified water, 112 mL of hydrogen gas at N.T.P. was collected at the catode in 965 seconds. The current passed, in ampere, is : **[JEE-MAINS (ONLINE) 2018]**

(1) 2.0 (2) 1.0 (3) 0.1 (4) 0.5

**EC0152**

**6.** When 9.65 ampere current was passed for 1.0 hour into nitrobenzene in acidic medium, the amount of p-aminophenol produced is :- **[JEE-MAINS (ONLINE) 2018]**

(1) 10.9 g (2) 98.1 g (3) 109.0 g (4) 9.81 g

**EC0153**

7. For the cell $Zn(s)\,|\,Zn^{2+}(aq)\,||\,M^{x+}(aq)\,|\,M(s)$, different half cells and their standard electrode potentials are given below : **[JEE-MAINS (ONLINE) 2019]**

| $M^{x+}(aq)/M(s)$ | $Au^{3+}(aq)/Au(s)$ | $Ag^+(aq)/Ag(s)$ | $Fe^{3+}(aq)/Fe^{2+}(aq)$ | $Fe^{2+}(aq)/Fe(s)$ |
|---|---|---|---|---|
| $E^{o}_{M^{x+}/M^{(s)}}$ | 1.40 | 0.80 | 0.77 | –0.44 |

If $E^{o}_{Zn^{2+}/Zn} = -0.76\,V$, which cathode will give a mximum value of $E^{o}_{cell}$ per electron transferred ?

(1) $Fe^{3+}$ / $Fe^{2+}$ (2) $Ag^+$ / Ag (3) $Au^{3+}$ / Au (4) $Fe^{2+}$ / Fe

**EC0154**

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**8.** If the standard electrode potential for a cell is 2 V at 300 K, the equilibrium constant (K) for the reaction

$Zn(s) + Cu^{2+}(aq) \rightleftharpoons Zn^{2+}(aq) + Cu(s)$ **[JEE-MAINS (ONLINE) 2019]**

at 300 K is approximately. **(R = 8 JK$^{-1}$ mol$^{-1}$ , F = 96000 C mol$^{-1}$)**

(1) $e^{160}$ (2) $e^{320}$ (3) $e^{-160}$ (4) $e^{-80}$

**EC0155**

**9.** Given the equilibrium constant : **[JEE-MAINS (ONLINE) 2019]**

$K_C$ of the reaction :

$Cu(s) + 2Ag^+(aq) \rightarrow Cu^{2+}(aq) + 2Ag(s)$ is $10 \times 10^{15}$, calculate the $E^0_{cell}$ of this reaction at 298 K.

$\left[2.303\frac{RT}{F} \text{ at } 298\,K = 0.059V\right]$

(1) 0.04736 V (2) 0.4736 V (3) 0.4736 mV (4) 0.04736 mV

**EC0156**

**10.** The standard electrode potential $E^{\ominus}$ and its temeprature coefficient $\left(\frac{dE^{\ominus}}{dT}\right)$ for a cell are 2V and $-5 \times 10^{-4}$ VK$^{-1}$ at 300 K respectively. The cell reaction is **[JEE-MAINS (ONLINE) 2019]**

$Zn(s) + Cu^{2+}(aq) \rightarrow Zn^{2+}(aq) + Cu(s)$

The standard reaction enthalpy $\left(\Delta_r H^{\ominus}\right)$ at 300 K in kJ mol$^{-1}$ is,

**[Use : R = 8JK$^{-1}$ mol$^{-1}$ and F = 96,000 Cmol$^{-1}$]**

(1) –412.8 (2) –384.0 (3) 206.4 (4) 192.0

**EC0157**

**11.** $\wedge_m^{\circ}$ for NaCl, HCl and NaA are 126.4, 425.9 and 100.5 S cm$^2$mol$^{-1}$, respectively. If the conductivity of 0.001 M HA is $5\times10^{-5}$ S cm$^{-1}$, degree of dissociation of HA is : **[JEE-MAINS (ONLINE) 2019]**

(1) 0.75 (2) 0.125 (3) 0.25 (4) 0.50

**EC0158**

**12.** Consider the following reduction processes :

$Zn^{2+} + 2e^- \rightarrow Zn(s)$; $E^\circ = -0.76$ V ; $Ca^{2+} + 2e^- \rightarrow Ca(s)$; $E^\circ = -2.87$ V

$Mg^{2+} + 2e^- \rightarrow Mg(s)$; $E^\circ = -2.36$ V ; $Ni^{2+} + 2e^- \rightarrow Ni(s)$; $E^\circ = -0.25$ V

The reducing power of the metals increases in the order : **[JEE-MAINS (ONLINE) 2019]**

(1) Ca < Zn < Mg < Ni (2) Ni < Zn < Mg < Ca

(3) Zn < Mg < Ni < Ca (4) Ca < Mg < Zn < Ni

**EC0159**

**13.** In the cell : **[JEE-MAINS (ONLINE) 2019]**

$Pt(s) \mid H_2(g, 1 \text{ bar} \mid HCl(aq) \mid AgCl(s) \mid Ag(s) \mid Pt(s)$

the cell potential is 0.92V when a $10^{-6}$ molal HCl solution is used. The standard electrode potential of (AgCl/Ag,Cl$^-$) electrode is :

$\left\{\text{given}, \frac{2.303RT}{F} = 0.06V \text{ at } 298K\right\}$

(1) 0.20 V (2) 0.76 V (3) 0.40 V (4) 0.94 V

**EC0160**

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**14.** The anodic half-cell of lead-acid battery is recharged using electricity of 0.05 Faraday. The amount of $PbSO_4$ electrolyzed in gm during the process is : **(Molar mass of $PbSO_4$ = 303 g mol$^{-1}$)**

(1) 22.8 (2) 15.2 **[JEE-MAINS (ONLINE) 2019]**

(3) 7.6 (4) 11.4

**EC0161**

**15.** Given that the standard potentials (E°) of $Cu^{2+}/Cu$ and $Cu^{+}/Cu$ are 0.34 V and 0.522 V respectively, the E° of $Cu^{2+}/Cu^{+}$ is : **[JEE-MAINS (ONLINE) 2020]**

(1) +0.158 V (2) 0.182 V (3) –0.182 V (4) –0.158 V

**EC0162**

**16.** The equation that is incorrect is - **[JEE-MAINS (ONLINE) 2020]**

(1) $\left(\Lambda_m^0\right)_{NaBr} - \left(\Lambda_m^0\right)_{NaI} = \left(\Lambda_m^0\right)_{KBr} - \left(\Lambda_m^0\right)_{NaBr}$

(2) $\left(\Lambda_m^0\right)_{H_2O} = \left(\Lambda_m^0\right)_{HCl} + \left(\Lambda_m^0\right)_{NaOH} - \left(\Lambda_m^0\right)_{NaCl}$

(3) $\left(\Lambda_m^0\right)_{KCl} - \left(\Lambda_m^0\right)_{NaCl} = \left(\Lambda_m^0\right)_{KBr} - \left(\Lambda_m^0\right)_{NaBr}$

(4) $\left(\Lambda_m^0\right)_{NaBr} - \left(\Lambda_m^0\right)_{NaCl} = \left(\Lambda_m^0\right)_{KBr} - \left(\Lambda_m^0\right)_{KCl}$

**EC0163**

**17.** What would be the electrode potential for the given half cell reaction at pH = 5 ?_____ **[JEE-MAINS (ONLINE) 2020]**

$2H_2O \rightarrow O_2 + 4H^{\oplus} + 4e^-$ ; $E^0_{red}$ = 1.23 V

**(R = 8.314 J mol$^{-1}$ K$^{-1}$ ; Temp = 298 K; oxygen under std. atm. pressure of 1 bar)**

**EC0164**

**18.** For an electrochemical cell **[JEE-MAINS (ONLINE) 2020]**

$Sn(s)|Sn^{2+}(aq,1M)||Pb^{2+}(aq,1M)|Pb(s)$

the ratio $\frac{[Sn^{2+}]}{[Pb^{2+}]}$ when this cell attains equilibrium is _____.

(Given $E^0_{Sn^{2+}|Sn} = -0.14V$ , $E^0_{Pb^{2+}|Pb} = -0.13V, \frac{2.303RT}{F} = 0.06$)

**EC0165**

**19.** 108 g of silver (molar mass 108 g mol$^{-1}$) is deposited at cathode from $AgNO_3$(aq) solution by a certain quantity of electricity. The volume (in L) of oxygen gas produced at 273 K and 1 bar pressure from water by the same quantity of electricity is _____. **[JEE-MAINS (ONLINE) 2020]**

**EC0166**

**20.** Amongst the following, the form of water with the lowest ionic conductance at 298 K is:

(1) distilled water **[JEE-MAINS (ONLINE) 2020]**

(2) water from a well

(3) saline water used for intravenous injection

(4) sea water

**EC0167**

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**21.** Consider the following cell reaction : **[JEE-MAINS 2021]**

$$Cd(s) + Hg_2SO_4(s) + \frac{9}{5}H_2O_{(l)} \rightleftharpoons CdSO_4.\frac{9}{5}H_2O_{(s)} + 2Hg_{(l)}$$

The value of $E^0_{cell}$ is 4.315 V at 25°C. If $\Delta H° = -825.2$ kJ mol$^{-1}$, the standard entropy change $\Delta S°$ in J K$^{-1}$ is _____. (Nearest integer)

[Given : Faraday constant = 96487 C mol$^{-1}$] **EC0290**

**22.** Match List-I with List-II **[JEE-MAINS 2021]**

| **List-I** (Parameter) | | **List-II** (Unit) | |
|---|---|---|---|
| (a) | Cell constant | (i) | S cm$^2$ mol$^{-1}$ |
| (b) | Molar conductivity | (ii) | Dimensionless |
| (c) | Conductivity | (iii) | m$^{-1}$ |
| (d) | Degree of dissociation of electrolyte | (iv) | $\Omega^{-1}$ m$^{-1}$ |

Choose the most appropriate answer from the options given below :

(1) (a)-(iii), (b)-(i), (c)-(iv), (d)-(ii) (2) (a)-(iii), (b)-(i), (c)-(ii), (d)-(iv)

(3) (a)-(i), (b)-(iv), (c)-(iii), (d)-(ii) (4) (a)-(ii), (b)-(i), (c)-(iii), (d)-(iv) **EC0291**

**23.** The resistance of a conductivity cell with cell constant 1.14 cm$^{-1}$, containing 0.001 M KCl at 298K is 1500 Ω. The molar conductivity of 0.001 M KCl solution at 298 K in S cm$^2$ mol$^{-1}$ is_____. (Integer answer) **[JEE-MAINS 2021]**

**EC0292**

**24.** Emf of the following cell at 298 K in V is $x \times 10^{-2}$. $Zn|Zn^{2+}$ (0.1 M)||$Ag^+$ (0.01 M)| Ag

The value of $x$ is _____. (Rounded off to the nearest integer) **[JEE-MAINS 2021]**

[Given : $E^0_{Zn^{2+}/Zn} = -0.76V; E^0_{Ag^+/Ag} = +0.80V; \frac{2.303RT}{F} = 0.059$] **EC0293**

**25.** Copper reduces $NO_3^-$ into NO and $NO_2$ depending upon the concentration of $HNO_3$ in solution. (Assuming fixed $[Cu^{2+}]$ and $P_{NO} = P_{NO_2}$), the $HNO_3$ concentration at which the thermodynamic tendency for reduction of $NO_3^-$ into NO and $NO_2$ by copper is same is $10^x$ M. The value of 2x is _____. (Rounded-off to the nearest integer) **[JEE-MAINS 2021]**

[Given, $E^o_{Cu^{2+}/Cu}$ = 0.34 V, $E^o_{NO_3^-/NO}$ = 0.96 V, $E^o_{NO_3^-/NO_2}$ = 0.79 V and at 298 K, $\frac{RT}{F}(2.303) = 0.059$]

**EC0294**

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**26.** The magnitude of the change in oxidising power of the $MnO_4^- / Mn^{2+}$ couple is x × $10^{-4}$ V, if the $H^+$ concentration is decreased from 1 M to $10^{-4}$ M at 25°C. (Assume concentration of $MnO_4^-$ and $Mn^{2+}$ to be same on change in $H^+$ concentration). The value of x is _____. **[JEE-MAINS 2021]**

(Rounded off to the nearest integer) **EC0295**

$\left[\text{Given} : \frac{2.303\ RT}{F} = 0.059\right]$

**27.** The molar conductivities at infinite dilution of barium chloride, sulphuric acid and hydrochloric acid are 280, 860 and 426 $Scm^2\ mol^{-1}$ respectively. The molar conductivity at **[JEE-MAINS 2021]** infinite dilution of barium sulphate is _____S $cm^2\ mol^{-1}$(Round off to the Nearest Integer).

**EC0296**

**28.** A KCl solution of conductivity 0.14 S $m^{-1}$ shows a resistance of 4.19 Ω in a conductivity cell. If the same cell is filled with an HCl solution, the resistance drops to 1.03 Ω. The conductivity of the HCl solution is____× $10^{-2}$ S $m^{-1}$. (Round off to the Nearest Integer). **[JEE-MAINS 2021]**

**EC0297**

**29.** Given below are two statements : **EC0298**

**Statement I :** The E° value of $Ce^{4+}$ / $Ce^{3+}$ is + 1.74 V.

**Statement II :** Ce is more stable in $Ce^{4+}$ state than $Ce^{3+}$ state. **[JEE-MAINS 2021]**

In the light of the above statements, choose the most appropriate answer from the options given below

(1) Both statement I and statement II are correct

(2) Statement I is incorrect but statement II is correct

(3) Both statement I and statement II are incorrect

(4) Statement I is correct but statement II is incorrect

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## EXERCISE # J-ADVANCED

**1.** Consider the following cell reaction : **[JEE 2011]**

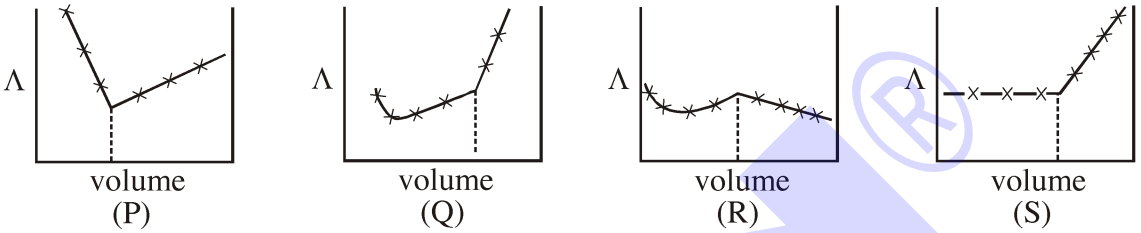
$2Fe_{(s)}+O_{2(g)}+4H^{+}_{(aq)} \rightarrow 2Fe^{2+}_{(a)}+2H_2O(\ell)$ ; $E^\circ = 1.67$ V

At$[Fe^{2+}] = 10^{-3}$ M, $P(O_2) = 0.1$ atm and pH = 3, the cell potential at 25°C is -

(A) 1.47 V (B) 1.77 V (C) 1.87 V (D) 1.57 V

**EC0178**

**2.** $AgNO_3$ (aq.) was added to an aqueous KCl solution gradually and the conductivity of the solution was measured. the plot of conductance ($\Lambda$) versus the volume of $AgNO_3$ is - **[JEE 2011]**




(A) (P) (B) (Q) (C) (R) (D) (S)

**EC0179**

***Paragraph for Question 3 and 4***

The electrochemical cell shown below is a concentration cell. **[JEE 2012]**

$M \mid M^{2+}$ (saturated solution of a sparingly soluble salt, $MX_2$) $\mid\mid M^{2+}$ (0.001 mol $dm^{-3}$) $\mid M$

The emf of the cell depends on the difference in concentrations of $M^{2+}$ ions at the two electrodes. The emf of the cell at 298 K is 0.059V.

**3.** The value of $\Delta G$ (kJ $mol^{-1}$) for the given cell is (take If = 96500 C $mol^{-1}$)

(A) –5.7 (B) 5.7 (C) 11.4 (D) –11.4

**EC0180**

**4.** The solubility product ($K_{sp}$; $mol^3\ dm^{-9}$) of $MX_2$ at 298 K based on the information available for the given concentration cell is (take $2.303 \times R \times 298|F = 0.059$ V)

(A) $1 \times 10^{-15}$ (B) $4 \times 10^{-15}$ (C) $1 \times 10^{-12}$ (D) $1 \times 10^{-12}$

**EC0181**

**5.** The standard reduction potential data at 25ºC is given below **[JEE-Adv. 2013]**

$E^o\ (Fe^{3+}, Fe^{2+}) = +0.77$ V ;

$E^o\ (Fe^{2+}, Fe) = -0.44$ V ;

$E^o\ (Cu^{2+}, Cu) = +0.34$ V ;

$E^o\ (Cu^{+}, Cu) = +0.52$ V ;

$E^o\ (O_2(g) + 4H^+ + 4e^- \rightarrow 2H_2O] = +1.23$ V ;

$E^o\ [(O_2(g) + 2H_2O + 4e^- \rightarrow 4OH^-)] = +0.40$ V ;

$E^o\ (Cr^{3+}, Cr) = -0.74$ V ;

$E^o\ (Cr^{2+}, Cr) = -0.91$ V ;

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Match $E^o$ of the redox pair in List-I with the values given in List-II and select the correct answer using the code given below the lists :

| | **List-I** | | **List-II** |
|---|---|---|---|
| (P) | $E^o(Fe^{3+}, Fe)$ | (1) | –0.18 V |
| (Q) | $E^o(4H_2O \rightleftharpoons 4H^+ + 4OH^+)$ | (2) | –0.4 V |
| (R) | $E^o(Cu^{2+} + Cu \rightarrow 2Cu^+)$ | (3) | –0.04 V |
| (S) | $E^o(Cr^{3+}, Cr^{2+})$ | (4) | –0·83 V |

**Codes :**

| | P | Q | R | S | | P | Q | R | S |
|---|---|---|---|---|---|---|---|---|---|
| (A) | 4 | 1 | 2 | 3 | (B) | 2 | 3 | 4 | 1 |
| (C) | 1 | 2 | 3 | 4 | (D) | 3 | 4 | 1 | 2 |

**EC0182**

**6.** An aqueous solution of X is added slowly to an aqueous solution of Y as shown in List-I. The variation in conductivity of these reactions is given in List-II. Match List-I with List-II and select the correct answer using the code given below the lists : **[JEE-Adv. 2013]**

| | **List-I** | | **List-II** |
|---|---|---|---|
| (P) | $(C2H5)3N + CH3COO_H$<br>X Y | (1) | Conductivity decreases and then increases |
| (Q) | $KI(0.1M) + AgNO3(0.01_M)$<br>X Y | (2) | Conductivity decreases and then does not change much |
| (R) | CH3COOH + KOH<br>X Y | (3) | Conductivity increases and then does not change much |
| (S) | NaOH + HI<br>X Y | (4) | Conductivity does not change much and then increases |

**Codes :**

| | P | Q | R | S | | P | Q | R | S |
|---|---|---|---|---|---|---|---|---|---|
| (A) | 3 | 4 | 2 | 1 | (B) | 4 | 3 | 2 | 1 |
| (C) | 2 | 3 | 4 | 1 | (D) | 1 | 4 | 3 | 2 |

**EC0183**

**7.** In a galvanic cell , the salt bridge - **[JEE-Adv. 2014]**

(A) Does not participate chemically in the cell reaction

(B) Stops the diffusion of ions from one electrode to another

(C) Is necessary for the occurence of the cell reaction

(D) Ensures mixing of the two electrolytic solutions

**EC0184**

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8. The molar conductivity of a solution of a weak acid HX (0.01 M) is 10 times smaller than the molar conductivity of a solution of a weak acid HY (0.1 M). If $\lambda^0_{X^-} \approx \lambda^0_{Y^-}$, the difference in their $pK_a$ values , $pK_a(HX) - pK_a(HY)$, is (consider degree of ionization of both acids to be <<1).

**[JEE-Adv. 2015]**

**EC0185**

9. All the energy released from the reaction $X \rightarrow Y$, $\Delta_r G^o = -193$ kJ $mol^{-1}$ is used for the oxidizing $M^+$ and $M^+ \rightarrow M^{3+} + 2e^-$, $E^o = -0.25$ V. **[JEE-Adv. 2015]**

Under standard conditions, the number of moles of $M^+$ oxidized when one mole of X is converted to Y is - [F = 96500 C $mol^{-1}$]

**EC0186**

10. For the following electrochemical cell at 298K, **[JEE-Adv. 2016]**

$Pt(s) \mid H_2(g, 1bar) \mid H^+(aq, 1M) \parallel M^{4+}(a), M^{2+}(a) \mid Pt(s)$

$E_{cell} = 0.092$ V when $\frac{[M^{2+}(aq.)]}{[M^{4+}(aq.)]} = 10^x$

Given : $E^0_{M^{4+}/M^{2+}} = 0.151V$ ; $2.303\frac{RT}{F} = 0.059V$

The value of x is -

(A) –2 (B) –1 (C) 1 (D) 2

**EC0187**

11. The conductance of a 0.0015 M aqueous solution of a weak monobasic acid was determined by using a conductivity cell consisting of platinized Pt electrodes. The distance between the electrodes is 120 cm with an area of cross section of 1 $cm^2$. The conductance of this solution was found to be $5 \times 10^{-7}$S. The pH of the solution is 4. The value of limiting molar conductivity $\left(\Lambda^0_m\right)$ of this weak monobasic acid in aqueous solution is $Z \times 10^2$ S $cm^2 mol^{-1}$. The value of Z is.

**EC0188**

12. For the following cell : **[JEE-Adv. 2017]**

$Zn(s) \mid ZnSO_4(aq.) \parallel CuSO_4(aq.) \mid Cu(s)$

when the concentration of $Zn^{2+}$ is 10 times the concentration of $Cu^{2+}$ , the expression for $\Delta G$ ( in J $mol^{-1}$) is

[F is Faraday constant , R is gas constant, T is temperature , $E^o$(cell) = 1.1V]

(A) 2.303 RT + 1.1F (B) 2.303 RT – 2.2F (C) 1.1 F (D) –2.2 F

**EC0189**

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**13.** Consider an electrochemical cell: A(s) | $A^{n+}$ (aq, 2M) || $B^{2n+}$ (aq, 1M) | B(s). The value of $\Delta H^{\theta}$ for the cell reaction is twice that of $\Delta G^{\theta}$ at 300 K. If the emf of the cell is zero, the $\Delta S^{\theta}$ (in $JK^{-1}$ $mol^{-1}$) of the cell reaction per mole of B formed at 300 K is___. **[JEE-Adv. 2018]**

(Given : ln (2) = 0.7, R (universal gas constant) = 8.3 J $K^{-1}$ $mol^{-1}$. H, S and G are enthalpy, entropy and Gibbs energy, respectively.) **EC0190**

**14.** For the electrochemical cell,

$$Mg(s) \mid Mg^{2+}(aq, 1\ M) \mid\mid Cu^{2+}(aq, 1M) \mid Cu(s)$$

the standard emf of the cell is 2.70 V at 300 K. When the concentration of $Mg^{2+}$ is changed to x M, the cell potential changes to 2.67 V at 300 K. The value of x is____.**[JEE-Adv. 2018]**

(Given, $\frac{F}{R}$ = 11500 $KV^{-1}$, where F is the Faraday constant and R is the gas constant, ln(10) = 2.30) **EC0191**

**15.** Consider a 70% efficient hydrogen-oxygen fuel cell working under standard conditions at 1 bar and 298 K. Its cell reaction is **[JEE-Adv. 2020]**

$$H_2(g) + \frac{1}{2}O_2(g) \rightarrow H_2O(\ell).$$

The work derived from the cell on the consumption of $1.0 \times 10^{-3}$ mol of $H_2(g)$ is used to compress 1.00 mol of a monoatomic ideal gas in a thermally insulted container. What is the change in the temperature (in K) of the ideal gas ? **EC0299**

The standard reduction potentials for the two half-cells are given below.

$O_2(g) + 4\ H^+(aq.) + 4e^- \rightarrow 2H_2O(\ell)$, $E^o = 1.23$ V,

$2H^+(aq.) + 2e^- \rightarrow H_2(g)$, $E^o = 0.00$V.

Use F = 96500 C $mol^{-1}$, R = 8.314 J $mol^{-1}K^{-1}$

**16.** Some standard electrode potentials at 298 K are given below: **[JEE-Adv. 2021]**

| | |
|---|---|
| $Pb^{2+}/Pb$ | -0.13 V |
| $Ni^{2+}/Ni$ | -0.24 V |
| $Cd^{2+}/Cd$ | -0.40 V |
| $Fe^{2+}/Fe$ | -0.44 V |

To a solution containing 0.001 M of $X^{2+}$ and 0.1 M of $Y^{2+}$, the metal rods X and Y are inserted (at 298 K) and connected by a conducting wire. This resulted in dissolution of X. The correct combination(s) of X and Y, respectively, is (are) **EC0300**

(Given: Gas constant, R = 8.314 J $K^{-1}$ $mol^{-1}$, Faraday constant, F = 96500 C $mol^{-1}$)

(A) Cd and Ni (B) Cd and Fe

(C) Ni and Pb (D) Ni and Fe

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**Paragraph for Q.17 and Q.18**

At 298 K, the limiting molar conductivity of a weak monobasic acid is $4 \times 10^2$ S $cm^2$ $mol^{-1}$. At 298 K, for an aqueous solution of the acid the degree of dissociation of $\alpha$ and the molar conductivity is $y \times 10^2$ $Scm^2$ $mol^{-1}$. At 298 K, upon 20 times dilution with water, the molar conductivity of the solution becomes $3y \times 10^2$ S $cm^2$ $mol^{-1}$. **[JEE-Adv. 2021]**

**17.** The value of $\alpha$ is _____. **EC0301**

**18.** The value of y is _____. **EC0302**

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# ANSWER KEY

## EXERCISE # S-I

1. (1.61)
2. (1.35)
3. (– 0.03 or –0.04)
4. (0.53 , 0.38)
5. (3)
6. (0.59)
7. (–322.00)
8. (–1303.00)
9. (110.00)
10. (E = –0.81)
11. (7)
12. (12.00)
13. (E = 1.15 or 1.16)
14. ($E^0_{cell}$ = +0.01, $E_{cell}$ = –0.07 or –0.08)
15. (2)
16. (0.80)
17. ($E^0$ = –0.14 or –0.15)
18. (–0.16)
19. (7.60)
20. (21.80)
21. (0.78)
22. (14.00)
23. (i) 107.00 (ii) 609.20
24. (3.86)
25. (0.02 or 0.03)
26. (4)
27. (0.05 or 0.06)
28. (0.41)
29. (a) 0.10 , (b) 0.03, (c) 0.30
30. (a) 0.75 , (b) 0.69 , (c)1.1
31. (i) 2.16 - 2.17 ; (ii) 1336. 15
32. (4)
33. (44.76 or 44.77, 12.53)
34. (3)
35. (0.11 )
36. (i) 54 (ii) 16.25
37. (1.02)
38. (115800, 347.40)
39. (7.50)
40. (114, 5926)
41. (60.00)
42. (1.83)
43. (2)
44. (i) 250.00 (ii) 125.00
45. (440.00)
46. (2500.00)
47. (i) 625.00 (ii) 1.60
48. (4)
49. (125.00)
50. (400.00)
51. (510.00)
52. (150.00)
53. (0.80)
54. (0.12 )
55. (0.50 , 3)
56. (9)
57. (i) 7 (ii) 14
58. (10.00)
59. (9.40)
60. (0.50)

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## EXERCISE # S-II

| | | | |
|---|---|---|---|
| 1. (– 0.45 or – 0.46) | 2. (i) 1.27 , (ii) 245.11 | 3. (–0.22) | 4. (16.00) |
| 5. (200.00) | 6. (–0.03 or –0.04) | 7. (7.00) | 8. (268.00) |
| 9. (9.70) | 10. (42.21) | 11. (1.85 or 1.86) | 12. (0.14 or 0.15) |
| 13. (0.04 ) | 14. (0.19) | 15. (2) | 16. (0.52 , 0.61) |
| 17. (0.02 or 0.03) | 18. (193.00) | 19. (9.60, 0.01) | 20. (93.65 or 93.66) |

## EXERCISE # O-I

| Que. | 1 | 2 | 3 | 4 | 5 | 6 | 7 | 8 | 9 | 10 |
|---|---|---|---|---|---|---|---|---|---|---|
| Ans. | B | C | A | C | D | C | A | C | A | A |
| Que. | 11 | 12 | 13 | 14 | 15 | 16 | 17 | 18 | 19 | 20 |
| Ans. | C | C | A | D | D | A | B | C | B | D |
| Que. | 21 | 22 | 23 | 24 | 25 | 26 | 27 | 28 | 29 | 30 |
| Ans. | C | A | A | D | B | A | C | C | A | B |
| Que. | 31 | 32 | 33 | 34 | 35 | 36 | 37 | 38 | 39 | 40 |
| Ans. | A | D | C | B | C | A | D | D | C | A |
| Que. | 41 | 42 | 43 | 44 | 45 | 46 | 47 | 48 | 49 | 50 |
| Ans. | C | C | B | C | B | D | C | A | C | C |
| Que. | 51 | 52 | 53 | 54 | 55 | 56 | 57 | 58 | 59 | 60 |
| Ans. | A | A | B | B | D | A | C | A | C | B |
| Que. | 61 | 62 | | | | | | | | |
| Ans. | C | D | | | | | | | | |

## EXERCISE # O-II

| Que. | 1 | 2 | 3 | 4 | 5 | 6 | 7 | 8 | 9 | 10 |
|---|---|---|---|---|---|---|---|---|---|---|
| Ans. | B | D | A | B | B | A | A | B | D | C |
| Que. | 11 | 12 | 13 | 14 | 15 | 16 | 17 | 18 | 19 | 20 |
| Ans. | A | C | B | D | D | B | C | C | B | C,D |
| Que. | 21 | 22 | 23 | 24 | 25 | 26 | 27 | 28 | 29 | 30 |
| Ans. | B,C,D | A,B | B,C | A,D | A | A | D | C | A | C |
| Que. | 31 | | | | 32 | | | | | |
| Ans. | A-P,Q | B-P,Q | C-Q,R | D-P,S | C | | | | | |

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## EXERCISE # J-MAINS

| Que. | 1 | 2 | 3 | 4 | 5 | 6 | 7 | 8 | 9 | 10 |
|---|---|---|---|---|---|---|---|---|---|---|
| Ans. | 1 | 3 | 1 | 2 | 2 | 4 | 2 | 1 | 2 | 1 |
| Que. | 11 | 12 | 13 | 14 | 15 | 16 | 17 | 18 | 19 | 20 |
| Ans. | 2 | 2 | 1 | 3 | 1 | 1 | -0.93 to -0.94 | 2.13 to 2.17 | 5.66 to 5.68 | 1 |
| Que. | 21 | 22 | 23 | 24 | 25 | 26 | 27 | 28 | 29 | |
| Ans. | 25 | 1 | 760 | 147 | 4 | 3776 | 288 | 57 | 4 | |

## EXERCISE # J-ADVANCED

| Que. | 1 | 2 | 3 | 4 | 5 | 6 | 7 | 8 | 9 | 10 |
|---|---|---|---|---|---|---|---|---|---|---|
| Ans. | D | D | D | B | D | A | A,B | 3 | 4 | D |
| Que. | 11 | 12 | 13 | 14 | 15 | 16 | 17 | 18 | | |
| Ans. | 6 | B | -11.62 | 10 | 13.32 | A,B,C | 0.21or0.22 | 0.86 | | |

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